

A linear bound for the Paulsen problem

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Abstract

We prove that every real ε -nearly equal-norm Parseval frame of n vectors in $\mathbb{R}^d$ is within squared Frobenius distance $C\varepsilon d$ of an equal-norm Parseval frame, where C is an absolute constant; this is optimal up to the value of C . Equivalently, every rank- d orthogonal projection on $\mathbb{R}^n$ whose diagonal entries deviate from d/n by at most β is within squared Frobenius distance $Cn\beta$ of a rank- d orthogonal projection with constant diagonal. The proof has two parts. A deterministic *static balancing theorem*, proved by minimising a log-determinant potential over a box, provides a row scaling that corrects every diagonal error that is small compared with the inverse of the *barrier constant*: a half-set hitting time of the random walk whose jump rates are the squared entries P_{ij}^2 of the projection. Two random *seeds* then make the barrier constant small at linear cost: a filtered Gaussian tangent perturbation with a deterministic drift when $d \gtrsim \log n$, and a row-wise tangent perturbation with row renormalisation when $n \gtrsim d^2$. Bounded d is handled by the Hamilton–Moitra argument. The main theorem has been formalised in Lean 4 with Mathlib.

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*AI system (gpt-astra).

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1 The result and the plan

1.1 Statement

A frame of n vectors in $\mathbb{R}^d$ is a matrix $X \in \mathbb{R}^{n \times d}$ whose i th row is x_i^T . Distances are *squared* Frobenius distances $\|X - Y\|_F^2 = \sum_i \|x_i - y_i\|^2$, with no normalisation. Throughout,

$$a = d/n.$$

X is *Parseval* if $X^T X = I_d$ and *equal-norm* if $\|x_i\|^2 = a$ for all i ; an equal-norm Parseval frame is an *ENP frame*. X is an ε -*nearly ENP frame* if

$$(1 - \varepsilon)I_d \preceq X^T X \preceq (1 + \varepsilon)I_d, \quad (1 - \varepsilon)a \leq \|x_i\|^2 \leq (1 + \varepsilon)a \quad (1 \leq i \leq n). \quad (1.1)$$

Theorem 1.1 (Linear Paulsen bound). *There is an absolute constant C such that for all $1 \leq d \leq n$, all $0 < \varepsilon < 1$ and every real ε -nearly ENP frame $X \in \mathbb{R}^{n \times d}$ there is an ENP frame $W \in \mathbb{R}^{n \times d}$ with $\|X - W\|_F^2 \leq C\varepsilon d$.*

The bound is optimal up to the constant (Remark 1.3). Kwok, Lau, Lee and Ramachandran [3] proved the first bound independent of n , $O(\varepsilon d^{13/2})$; Hamilton and Moitra [2] improved it to $O(\varepsilon d^2)$; Lau and Ramachandran announced the linear bound [4], via a different route. Appendix B describes a Lean 4 formalisation of Theorems 1.1 and 1.2.

For a Parseval U write $P = UU^T$ for the orthogonal projection onto its column space; then $P_{ii} = \|u_i\|^2$. The problem has an equivalent form in which the error is additive.

Theorem 1.2 (Projection form). *There is an absolute constant C such that for every $n \geq 1$ and every rank- d orthogonal projection P on $\mathbb{R}^n$, with $\beta = \max_i |P_{ii} - d/n|$, there is a rank- d orthogonal projection Q with $Q_{ii} = d/n$ for all i and $\|P - Q\|_F^2 \leq Cn\beta$.*

This reformulation and the complement symmetry are due to Cahill and Casazza [1]: both β and $\|P - Q\|_F$ are unchanged when (P, Q) is replaced by $(I - P, I - Q)$, so in Theorem 1.2 one may assume $d \leq n/2$. In Section 6, Theorem 1.2 is derived from an equal-norm formulation (Proposition 6.1) and Theorem 1.1 from Theorem 1.2.

Remark 1.3 (Optimality). Let $d = 2d_1$, $n = 2n_1$, and $1 \leq k \leq n_1/4$ with $n_1 - k \geq d_1$; put $\varepsilon = 4k/n$. Let $X = X_1 \oplus X_2$, where X_1 is an ENP frame of $n_1 - k$ vectors in $\mathbb{R}^{d_1}$ and X_2 one of $n_1 + k$ vectors in $\mathbb{R}^{d_1}$ (Corollary 3.2). Then X is Parseval, and its squared row norms $a(1 \mp 2k/n)^{-1}$ make it an ε -nearly ENP frame. Let W be ENP, and let E and A be the principal blocks of XX^T and WW^T on the first $n_1 - k$ rows. Then E is a projection of trace d_1 and A a positive contraction of trace $(n_1 - k)a$, so $\Delta = E - A$ has $\text{tr } \Delta = ka = \varepsilon d/4$. Since $\text{tr}(E\Delta) = \text{tr } \Delta + \text{tr}((I - E)A) \geq \text{tr } \Delta$,

$$\text{tr}(A - A^2) = -\text{tr } \Delta + 2\text{tr}(E\Delta) - \text{tr } \Delta^2 \geq \text{tr } \Delta - \|\Delta\|_F^2.$$

As WW^T is a projection, its off-diagonal block has squared norm $\text{tr}(A - A^2)$, while that of XX^T vanishes; hence $\|XX^T - WW^T\|_F^2 \geq \|\Delta\|_F^2 + 2(\text{tr } \Delta - \|\Delta\|_F^2) \geq 2\text{tr } \Delta - \|XX^T - WW^T\|_F^2$, i.e. $\|XX^T - WW^T\|_F^2 \geq \text{tr } \Delta$, and by Lemma 2.1 $\|X - W\|_F^2 \geq \frac{1}{2}\text{tr } \Delta = \varepsilon d/8$.

Conventions. c, C denote positive absolute constants whose value may change between occurrences; constants with subscripts or names are fixed once chosen. $\|\cdot\|_{\text{op}}$ is the Euclidean operator norm, $\mathbf{1}$ the all-ones vector, $\text{Diag}(x)$ the diagonal matrix with diagonal x , $\text{diag}(M)$ the diagonal of M as a vector, and $\Pi_{\mathcal{S}}$ the orthogonal projection onto a subspace $\mathcal{S}$. $A \preceq B$ is the Loewner order. All Gaussian vectors are real and centred.

1.2 The plan

The engine: scaling. Let U be Parseval with projection P and diagonal $p = \text{diag } P$. For $s \in \mathbb{R}^n$ the matrix $\text{Diag}(e^s)U$ has a column space with projection $P(s)$, and $s \mapsto \text{diag } P(s)$ is the gradient of the convex function $\frac{1}{2} \log \det(U^T \text{Diag}(e^{2s})U)$, whose Hessian at $s = 0$ is $2L_P$, where

$$L_P = \text{Diag}(p) - P \circ P$$

is the graph Laplacian with edge weights P_{ij}^2 (Lemma 2.2). To first order, changing the diagonal by $\delta \perp \mathbf{1}$ therefore requires $s \approx \frac{1}{2}L_P^+ \delta$, at a cost of about $s^T L_P s \approx \frac{1}{4}\delta^T L_P^+ \delta$. If L_P has spectral gap λ and $|\delta_i| \leq \varepsilon a$, the cost is at most $n\varepsilon^2 a^2/\lambda$. For a “random-looking” P one has $\lambda \approx a$ and the cost is $\varepsilon^2 d$, but L_P can be arbitrarily badly connected: for an exact direct sum the components cannot exchange diagonal mass at all.

Perturb, then scale. Perturb U by a random horizontal tangent of size $t = \sqrt{K_0}\varepsilon$, with K_0 a large constant, at cost $O(t^2 d) = O(\varepsilon d)$. Generic entries of the new projection then have size $t\sqrt{a/n}$, so most edges acquire weight $\gtrsim at^2/n$, and the random walk with rates P_{ij}^2 reaches any half of the vertices in time $O(1/(at^2))$: the *barrier constant* (Definition 2.6 and Lemma 2.10) is $O(1/(at^2))$. The static balancing theorem (Theorem 2.8) then removes exactly every diagonal error β with $\beta \cdot O(1/(at^2)) \leq \frac{1}{2}$, at cost $\frac{15}{4}n\beta = O(t^2 d)$. The ℓ_∞ (rather than spectral) formulation is essential: the scaling must stay in a box $\|s\|_\infty \leq 1$, and an ℓ_2 argument loses a factor $\sqrt{n}$.

The difficulty. The perturbed frame must therefore have diagonal error at most θat^2 in ℓ_∞ , for a small fixed θ . The initial error $\varepsilon a = at^2/K_0$ is acceptable. But a random tangent of size t moves each diagonal entry *linearly* by about $t\sqrt{a/n}$, which exceeds at^2 as soon as $\varepsilon \lesssim 1/d$. Two different constructions deal with this.

Seed 1: many rows ($n \geq Bd^2$, [Section 4](#)). Here the input is an equal-norm frame X with spectral error $\eta = \|X^T X - I\|_{\text{op}}$, and t^2 is a large multiple of η . Perturb each row *orthogonally to itself* and renormalise. Row norms stay exact, but the frame operator moves. Its first-order change is removed by conditioning the Gaussian on a subspace of codimension at most d^2 , and its second-order fluctuation is an average over n rows, of size $t^2 d / \sqrt{n}$. The spectral error of the perturbed frame is therefore $\eta + O(t^2 d / \sqrt{n} + t^2 d^2 / n + t^3)$, which is at most θt^2 once t^2 / η and B are large and t is small; whitening turns a spectral error δ into a diagonal error at most $2a\delta$. The cubic term is $O(t^3)$, not the termwise $O(t^3 d)$, because the first-order constraint allows the normalisation weights to be centred ([Lemma 4.4](#)); this is why the seed works for every $\eta \leq c_*$, with no relation required between η and d .

Seed 2: moderate rows ($d \geq A \log 2n$, [Section 5](#)). Perturb U in the horizontal tangent space of the Grassmannian, so that the frame stays exactly Parseval, with a Gaussian whose covariance is filtered: $C_\rho = \rho(I - \Omega)(\rho I + \Omega)^{-1}$, where Ω is the normalised Gram operator of the linear diagonal map $\mathcal{A}$ and $\rho = D^2 t^2$ for a large constant D . The filter makes the linear diagonal change $O(t\sqrt{\rho a \log n / n}) = O(Dat^2 \sqrt{\log n / d})$, negligible when $d \gg \log n$, while leaving the weak (connecting) directions untouched. The price is a second-order bias $t^2 b_*$, which is not small in ℓ_∞ . A commutator identity shows that $b_* = \mathcal{A}m_*$ for an explicit deterministic tangent direction m_* with $\|m_*\|_{\text{F}}^2 \leq 2a/\rho$ ([Lemmas 5.5](#) and [5.6](#)). We add the drift $\frac{t}{2}m_*$ to the noise: its first-order effect cancels the bias exactly, and its other effects are $O(at^2/D)$. All remaining errors are at most θat^2 by concentration, which is where $d \geq A \log 2n$ is used. Only a bounded, deterministic set of rows needs a spectral argument, and second moments suffice there.

Assembly ([Section 6](#)). After complementation and normalisation it suffices to treat equal-norm frames with $2d \leq n$ and spectral error η . If $d < d_0$ (a large absolute constant), the Hamilton–Moitra bound $\eta d(d-1) \leq d_0 \eta d$ suffices ([Section 3](#)). If $d \geq d_0$ and $n \geq Bd^2$, Seed 1 applies. If $d \geq d_0$ and $n < Bd^2$, then $A \log 2n \leq A \log(2Bd^2) \leq d$ and Seed 2 applies.

regime	construction	where
$d < d_0$	radial isotropic position, ordered coordinates (Hamilton–Moitra)	Section 3
$d \geq d_0, n \geq Bd^2$	row-tangent Gaussian, row normalisation, whitening, balancing	Section 4
$d \geq d_0, 2d \leq n < Bd^2$	filtered horizontal Gaussian with drift, balancing	Section 5
$d > n/2$	complement	Section 6

2 The scaling toolbox

Throughout this section $U \in \mathbb{R}^{n \times d}$ is Parseval, $P = UU^T$, $p = \text{diag } P$, and $L_P = \text{Diag}(p) - P \circ P$. For nonnegative symmetric weights w_{ij} on $[n]$ the weighted Laplacian L acts by $(Lx)_i = \sum_j w_{ij}(x_i - x_j)$; diagonal weights do not contribute. Unless another Laplacian is specified, $L = L_P$. We write $\text{osc}(x) = \max_i x_i - \min_i x_i$.

2.1 Projections, distances, and the squared-Gram Laplacian

Lemma 2.1 (Frames versus projections). *Let $U, W \in \mathbb{R}^{n \times d}$ be Parseval with projections P, Q , and let $\sigma_1, \dots, \sigma_d \in [0, 1]$ be the singular values of $U^T W$. Then*

$$\min_{R \in O(d)} \|U - WR\|_{\text{F}}^2 = 2 \sum_k (1 - \sigma_k) \leq \|P - Q\|_{\text{F}}^2 = 2 \sum_k (1 - \sigma_k^2) = 2\|(I - P)W\|_{\text{F}}^2 \leq 2\|U - W\|_{\text{F}}^2.$$

Right multiplication by $R \in O(d)$ preserves Parsevalness and every row norm. If $X \in \mathbb{R}^{n \times d}$ has $\delta = \|X^T X - I\|_{\text{op}} < 1$, its polar factor $\tilde{X} = X(X^T X)^{-1/2}$ is Parseval, $\|X - \tilde{X}\|_{\text{F}}^2 \leq d\delta^2$, and $\|x_i\|^2/(1+\delta) \leq \|\tilde{x}_i\|^2 \leq \|x_i\|^2/(1-\delta)$. If moreover $\delta \leq \frac{1}{2}$, then $|\|\tilde{x}_i\|^2 - \|x_i\|^2| \leq 2\delta\|x_i\|^2$ and row i of $\tilde{X}\tilde{X}^T - XX^T$ has squared norm at most $6\delta^2\|x_i\|^2$.

Proof. $\|U - WR\|_{\text{F}}^2 = 2d - 2\text{tr}(U^T WR)$, maximised over R by a polar factor of $U^T W$ (any orthogonal R with $U^T WR \succeq 0$; it need not be unique), giving $2d - 2\sum_k \sigma_k$. Next $\|P - Q\|_{\text{F}}^2 = 2d - 2\text{tr}(PQ) = 2d - 2\|U^T W\|_{\text{F}}^2$ and $\|(I - P)W\|_{\text{F}}^2 = d - \|U^T W\|_{\text{F}}^2$; also $1 - \sigma \leq 1 - \sigma^2$ on $[0, 1]$, and $(I - P)W = (I - P)(W - U)$ gives $\|(I - P)W\|_{\text{F}} \leq \|U - W\|_{\text{F}}$. For the polar factor, if ς_k are the singular values of X then $\|X - \tilde{X}\|_{\text{F}}^2 = \sum_k (\varsigma_k - 1)^2 \leq \sum_k (\varsigma_k^2 - 1)^2$, and $\|\tilde{x}_i\|^2 = x_i^T (X^T X)^{-1} x_i$. With $E = (X^T X)^{-1} - I$, so $\|E\|_{\text{op}} \leq \delta/(1 - \delta) \leq 2\delta$, row i of $\tilde{X}\tilde{X}^T - XX^T = XEX^T$ is $XE x_i$, of squared norm at most $(1 + \delta)4\delta^2\|x_i\|^2$. $\square$

Lemma 2.2 (Energy identity). L is the Laplacian of the weights P_{ij}^2 ($i \neq j$), $L \succeq 0$, $L\mathbf{1} = 0$, $L_{I-P} = L_P$, and for $x \in \mathbb{R}^n$, $X = \text{Diag}(x)$,

$$x^T Lx = \frac{1}{2} \sum_{i,j} P_{ij}^2 (x_i - x_j)^2 = \|(I - P)XU\|_{\text{F}}^2 = \frac{1}{2} \| [X, P] \|_{\text{F}}^2.$$

Proof. $\sum_j P_{ij}^2 = (P^2)_{ii} = p_i$, so the rows of $\text{Diag}(p) - P \circ P$ sum to zero and its off-diagonal entries are $-P_{ij}^2$. Next $\|(I - P)XU\|_{\text{F}}^2 = \text{tr}(XPX(I - P)) = \sum_i x_i^2 p_i - \sum_{i,j} x_i x_j P_{ij}^2$, and $[X, P]_{ij} = (x_i - x_j)P_{ij}$. For $I - P$ the off-diagonal weights are again P_{ij}^2 . $\square$

Lemma 2.3 (Scaled projections). For $w \in \mathbb{R}_{>0}^n$ put $D = \text{Diag}(w)$, $M = (U^T D^2 U)^{-1/2}$ and $W = DUM$. Then W is Parseval with column space $D \text{im } U$, and

$$\|WW^T - P\|_{\text{F}}^2 = 2\|(I - P)W\|_{\text{F}}^2 \leq \frac{2w^T Lw}{\min_i w_i^2}.$$

Proof. $W^T W = MU^T D^2 U M = I$ and $U^T D^2 U \succeq (\min_i w_i)^2 I$, so $\|M\|_{\text{op}} \leq 1/\min_i w_i$. By Lemmas 2.1 and 2.2, $\|(I - P)W\|_{\text{F}}^2 \leq \|M\|_{\text{op}}^2 \|(I - P)DU\|_{\text{F}}^2 = \|M\|_{\text{op}}^2 w^T Lw$. $\square$

2.2 Diagonal scaling

For $s \in \mathbb{R}^n$ let $D_s = \text{Diag}(e^s)$, let $P(s)$ be the projection onto $D_s \text{im } U$, $r(s) = \text{diag } P(s)$, and $F(s) = \frac{1}{2} \log \det(U^T D_s^2 U)$.

Lemma 2.4. F is smooth and convex, $\nabla F(s) = r(s)$, $\sum_i r_i(s) = d$, and $F(s + c\mathbf{1}) = F(s) + cd$.

Proof. With $G = U^T D_s^2 U = \sum_i e^{2s_i} u_i u_i^T$ one has $\partial_i F = e^{2s_i} u_i^T G^{-1} u_i = P(s)_{ii}$, because $P(s) = D_s U G^{-1} U^T D_s$; differentiating again gives $\nabla^2 F = 2L_{P(s)} \succeq 0$. The trace of $P(s)$ is d , and G scales by e^{2c} under $s \mapsto s + c\mathbf{1}$. $\square$

Lemma 2.5 (One-sided scaling inequalities). For $z \in \mathbb{R}_{>0}^n$ let r be the diagonal of the projection onto $\text{Diag}(\sqrt{z}) \text{im } U$. Then, coordinatewise,

$$Lz \leq z \circ (r - p), \quad L(z^{-1}) \leq -z^{-1} \circ (r - p).$$

Proof. Let $G = U^T \text{Diag}(z)U$. Since $G + z_i^2 G^{-1} - 2z_i I = (G^{1/2} - z_i G^{-1/2})^2 \succeq 0$, evaluating at u_i and using $u_i^T G u_i = \sum_j P_{ij}^2 z_j = p_i z_i - (Lz)_i$ and $z_i^2 u_i^T G^{-1} u_i = z_i r_i$ gives the first inequality. Let $V \in \mathbb{R}^{n \times (n-d)}$ be an isometry onto $(\text{im } U)^\perp$. The subspaces $\text{Diag}(\sqrt{z}) \text{im } U$ and $\text{Diag}(z^{-1/2}) \text{im } V$ are orthogonal complements, so the projection onto the latter has diagonal $\mathbf{1} - r$, while $VV^T = I - P$ has diagonal $\mathbf{1} - p$ and Laplacian L (Lemma 2.2). The first inequality for (V, z^{-1}) is the second. (If $d = n$ then $L = 0$ and $r = p = \mathbf{1}$.) $\square$

2.3 The barrier constant and static balancing

Definition 2.6 (Barrier constant). Let L be a weighted Laplacian on $[n]$. For $S \subseteq [n]$, an S -barrier is a vector $\psi \geq 0$ with $(L\psi)_i \geq 1$ for every $i \notin S$. The barrier constant $H(L) \in (0, \infty]$ is the least H such that every S with $|S| \geq n/2$ has an S -barrier with $\|\psi\|_\infty \leq H$ ($H(L) = \infty$ if some such S has no barrier).

For the continuous-time random walk that jumps from i to j at rate w_{ij} , the expected hitting time $h_S(x) = \mathbb{E}_x \tau_S$ of S vanishes on S and satisfies $(Lh_S)_i = 1$ off S . If ψ is any S -barrier and $\beta' > 1$, the maximum principle argument in the proof of Lemma 2.7, applied to $h_S - \beta'\psi$, gives $h_S \leq \beta'\psi$. So h_S is the minimal S -barrier, and $H(L_P)$ is the half-set hitting time of the walk with rates P_{ij}^2 : the largest expected time, over all starting points and all S with $|S| \geq n/2$, needed to reach S . If $H(L) < \infty$ then the graph is connected: for a component J with $|J| \leq n/2$, summing $(L\psi)_i$ over $i \in J$ gives 0, so $S = [n] \setminus J$ has no barrier. The barrier constant is within a factor 2 of the ℓ_∞ Poisson constant of L (Remark 2.11). A small barrier constant controls scalings in ℓ_∞ :

Lemma 2.7 (Median barrier). Let L be a weighted Laplacian on $[n]$ with $H(L) \leq H < \infty$, let $\beta \geq 0$ with $\beta H < 1$, let $z \in \mathbb{R}_{>0}^n$, and let m be a median of z (so that $\{z \leq m\}$ and $\{z \geq m\}$ both have at least $n/2$ elements). If

$$(Lz)_i \leq \beta z_i \text{ when } z_i > m, \quad (Lz^{-1})_i \leq \beta z_i^{-1} \text{ when } z_i < m,$$

then $(1 - \beta H) \max_i z_i \leq m \leq (1 - \beta H)^{-1} \min_i z_i$.

Proof. Let $S = \{z \leq m\}$, let ψ be an S -barrier with $\|\psi\|_\infty \leq H$, let $M = \max z$ and $\beta' > \beta$. The vector $v = z - m\mathbf{1} - \beta'M\psi$ is ≤ 0 on S , and for $i \notin S$, $(Lv)_i \leq \beta z_i - \beta'M < 0$. At a maximiser i of v , $(Lv)_i = \sum_j w_{ij}(v_i - v_j) \geq 0$; so every maximiser lies in S , and $v \leq 0$. Hence $M \leq m + \beta'MH$; let $\beta' \downarrow \beta$. Apply the same argument to z^{-1} , whose median is m^{-1} , with $S = \{z \geq m\}$. $\square$

Theorem 2.8 (Static balancing). Suppose $H(L_P) \leq H < \infty$, and $q \in \mathbb{R}^n$ satisfies $\sum_i q_i = d$ and $\beta H \leq \frac{1}{2}$, where $\beta = \|q - p\|_\infty$. Then there is a Parseval W with $\text{diag}(WW^T) = q$ and

$$\|U - W\|_F^2 \leq \|UU^T - WW^T\|_F^2 \leq \frac{15}{4} \|q - p\|_1.$$

W is obtained from U by a row scaling with ratio of scales at most 2, whitening, and an orthogonal right factor.

Proof. Let s minimise $\Phi(s) = F(s) - \langle q, s \rangle$ over the cube $[0, 1]^n$ (compactness), and write $r = r(s)$. Since $\partial_i \Phi = r_i - q_i$ (Lemma 2.4), coordinatewise optimality gives

$$r_i \leq q_i \text{ if } s_i > 0, \quad r_i \geq q_i \text{ if } s_i < 1. \quad (2.1)$$

Let $z = e^{2s}$ with median m , so $1 \leq \min z \leq m \leq \max z \leq e^2$. If $z_i > m$ then $s_i > 0$, so by Lemma 2.5 and (2.1) $(Lz)_i \leq z_i(r_i - p_i) \leq z_i(q_i - p_i) \leq \beta z_i$; if $z_i < m$ then $s_i < 1$ and likewise $(Lz^{-1})_i \leq \beta z_i^{-1}$. By Lemma 2.7, $\max z / \min z \leq (1 - \beta H)^{-2} \leq 4$, i.e. $\text{osc}(s) \leq \log 2 < 1$. Hence s does not meet both faces of the cube: either all $s_i > 0$, and then $r \leq q$, or some $s_j = 0$, all $s_i < 1$, and then $r \geq q$. Since $\sum r_i = d = \sum q_i$, in both cases $r = q$.

For the distance let $w = e^s$, $f = q - p$, and let c be the midpoint of the range of z^2 . Lemma 2.5 with $r = q$ gives $Lz \leq z \circ f$, so, using $\sum_i f_i = 0$,

$$z^T Lz \leq \sum_i z_i^2 f_i = \sum_i (z_i^2 - c) f_i \leq \frac{1}{2} \text{osc}(z^2) \|f\|_1.$$

As $(z_i - z_j)^2 \geq 4 \min w^2 (w_i - w_j)^2$, [Lemmas 2.2](#) and [2.3](#) give

$$\|P(s) - P\|_F^2 \leq \frac{2w^T Lw}{\min w^2} \leq \frac{z^T Lz}{2 \min z^2} \leq \frac{\text{osc}(z^2)}{4 \min z^2} \|f\|_1 \leq \frac{4^2 - 1}{4} \|f\|_1.$$

Finally apply [Lemma 2.1](#) to the whitened frame $DU(U^T D^2 U)^{-1/2}$, $D = \text{Diag}(w)$, whose projection is $P(s)$. $\square$

Remark 2.9. The hypothesis forces $q_i \geq p_i/2 > 0$: for $n \geq 2$, the barrier for $S = [n] \setminus \{i\}$ gives $1 \leq (L\psi)_i \leq p_i(1 - p_i)H$, so $\beta \leq 1/(2H) \leq p_i/2$. The proof uses no existence theorem for scalings, no flow, and no genericity or connectivity argument: the box supplies a minimiser, the median barrier keeps it away from one face, and the trace identity $\sum_i r_i = d$ does the rest. With $\beta H < 1$ and a box of side $T > -\log(1 - \beta H)$ the same proof gives scale ratio $(1 - \beta H)^{-1}$ and cost $\frac{1}{4}((1 - \beta H)^{-4} - 1)\|q - p\|_1$.

2.4 Barriers from a dense core

Lemma 2.10 (Dense core with exceptional vertices). *Let L be the Laplacian of nonnegative symmetric weights w_{ij} on $[n]$, and let $[n] = \mathcal{G} \sqcup \mathcal{B}$. Suppose:*

- (a) *there is $\kappa > 0$ with $\sum_{j \in S} w_{ij} \geq \kappa$ for every $i \in \mathcal{G}$ and every $S \subseteq [n] \setminus \{i\}$ with $|S| \geq n/2$;*
- (b) *there are $y \in \mathbb{R}^{\mathcal{B}}$, $y \geq 0$, and $R, F_{\mathcal{B}} \geq 0$ with $\|y\|_{\infty} \leq R$, $(L_{\mathcal{B}\mathcal{B}}y)_i \geq 1$ for $i \in \mathcal{B}$, and $\sum_{j \in \mathcal{B}} w_{ij}y_j \leq F_{\mathcal{B}}$ for $i \in \mathcal{G}$. (If $\mathcal{B} = \emptyset$ take $R = F_{\mathcal{B}} = 0$.)*

Then $H(L) \leq (1 + F_{\mathcal{B}})/\kappa + R$. Moreover, with $b = |\mathcal{B}|$:

- (i) *(a) holds with $\kappa = \vartheta\gamma$ if $\vartheta \in (0, \frac{1}{2}]$, $\gamma > 0$, and every $i \in \mathcal{G}$ has at least $(\frac{1}{2} + \vartheta)n$ indices $j \neq i$ with $w_{ij} \geq \gamma/n$;*
- (ii) *(b) holds with $R = \sqrt{b}/\lambda$, $F_{\mathcal{B}} = b$, if $x^T Lx \geq \lambda\|x\|^2$ for every x supported on $\mathcal{B}$ ($\lambda > 0$);*
- (iii) *for $w_{ij} = P_{ij}^2$, (b) holds with $R = \frac{8}{3\alpha}$, $F_{\mathcal{B}} = \frac{2}{3\alpha} \max_i p_i$ if $\alpha > 0$, $p_i \geq \alpha/2$ on $\mathcal{B}$ and $\sum_{i \in \mathcal{B}} p_i \leq \frac{1}{4}$.*

Proof. Let $|S| \geq n/2$, put $\lambda_0 = (1 + F_{\mathcal{B}})/\kappa$, extend y by 0 to $\mathcal{G}$, and let $\psi = (\lambda_0 \mathbf{1} + y) \circ \mathbf{1}_{S^c}$, so $0 \leq \psi \leq \lambda_0 + R$ and $\psi_j \leq \lambda_0 + y_j$ for all j . For $i \in \mathcal{G} \setminus S$, $\psi_i = \lambda_0$ and $\psi_j = 0$ on S , so

$$(L\psi)_i \geq \lambda_0 \sum_{j \in S} w_{ij} - \sum_{j \in \mathcal{B}} w_{ij}y_j \geq \lambda_0 \kappa - F_{\mathcal{B}} = 1.$$

For $i \in \mathcal{B} \setminus S$, $\psi_i = \lambda_0 + y_i$, so $(L\psi)_i \geq \sum_j w_{ij}(y_i - y_j) = (L_{\mathcal{B}\mathcal{B}}y)_i \geq 1$. Thus ψ is an S -barrier.

(i) If N_i is the set of such j and $S \subseteq [n] \setminus \{i\}$, then $|N_i \cap S| \geq |N_i| + |S| - (n - 1) > \vartheta n$. (ii) $L = L_{\mathcal{B}\mathcal{B}} \succeq \lambda I$; let $y = L^{-1} \mathbf{1}_{\mathcal{B}}$. If y had a negative minimum at i , then $(Ly)_i = y_i \sum_{j \notin \mathcal{B}} w_{ij} + \sum_{j \in \mathcal{B}} w_{ij}(y_i - y_j) \leq 0$, a contradiction; so $y \geq 0$, and $\|y\|_{\infty} \leq \|y\|_2 \leq \sqrt{b}/\lambda$. Since $(L \mathbf{1}_{\mathcal{B}})_j = \sum_{i \in \mathcal{G}} w_{ij}$ for $j \in \mathcal{B}$, for each $i_0 \in \mathcal{G}$

$$\sum_{j \in \mathcal{B}} w_{i_0 j} y_j \leq \sum_{i \in \mathcal{G}} \sum_{j \in \mathcal{B}} w_{ij} y_j = \mathbf{1}_{\mathcal{B}}^T L y = b.$$

(iii) $P_{ij}^2 \leq p_i p_j$ gives, for $i \in \mathcal{B}$, $(L_{\mathcal{B}\mathcal{B}} \mathbf{1}_{\mathcal{B}})_i = p_i - \sum_{j \in \mathcal{B}} P_{ij}^2 \geq p_i(1 - \sum_{j \in \mathcal{B}} p_j) \geq \frac{3\alpha}{8}$, and for $i \in \mathcal{G}$, $\sum_{j \in \mathcal{B}} P_{ij}^2 \leq p_i/4$. Take $y = \frac{8}{3\alpha} \mathbf{1}_{\mathcal{B}}$. $\square$

Remark 2.11 (Poisson constants). The ℓ_∞ Poisson constant K of L is the least K such that every $f \perp \mathbf{1}$ has a solution of $Lg = f$ with $\|g\|_\infty \leq K\|f\|_\infty$. Then $K \leq H \leq 2K$. Indeed, for $|S| \geq n/2$ the forcing $f = \mathbf{1} - \frac{n}{|S|}\mathbf{1}_S$ has $\|f\|_\infty \leq 1$, and $g + K\mathbf{1}$ is an S -barrier; conversely, if $H < \infty$ the graph is connected, so $Lg = f$ is solvable for every $f \perp \mathbf{1}$, and if $\|f\|_\infty \leq 1$ and m is a median of g , the argument of [Lemma 2.7](#) applied to $g - m\mathbf{1} - \beta'\psi$ ($\beta' > 1$) gives $g \leq m + H$, and symmetrically $g \geq m - H$. Barriers are more convenient than Poisson solutions because they can be written down explicitly, as in [Lemma 2.10](#).

2.5 From a seed to an exact frame

Both constructions produce the following object.

Definition 2.12 (Seed). Let $\Theta \geq 2$ and $t > 0$. A (Θ, t) -seed for $X \in \mathbb{R}^{n \times d}$ is a Parseval $V \in \mathbb{R}^{n \times d}$ with

$$\|V - X\|_F^2 \leq \Theta t^2 d, \quad H(L_{VV^T}) \leq \frac{\Theta}{at^2}, \quad \max_i |(VV^T)_{ii} - a| \leq \frac{at^2}{2\Theta}.$$

Corollary 2.13. *If X has a (Θ, t) -seed, there is an ENP frame W with $\|X - W\|_F^2 \leq 3\Theta t^2 d$.*

Proof. Apply [Theorem 2.8](#) to V with $q = a\mathbf{1}$: $\beta H \leq \frac{1}{2}$, and $\|V - W\|_F^2 \leq \frac{15}{4}n \cdot \frac{at^2}{2\Theta} \leq t^2 d$. Then $\|X - W\|_F^2 \leq 2\|X - V\|_F^2 + 2\|V - W\|_F^2 \leq (2\Theta + 2)t^2 d \leq 3\Theta t^2 d$. $\square$

3 Bounded rank: the Hamilton–Moitra estimate

This section reproduces the radial-isotropic and ordered-coordinate argument of Hamilton and Moitra [\[2\]](#), with a direct compactness proof of the existence of the radial scaling. It is used for d below a large absolute constant, and it also shows that ENP frames exist.

Lemma 3.1. *Let $1 \leq d \leq n$, let $X \in \mathbb{R}^{n \times d}$ have $\|x_i\|^2 = a$ for all i , and let $\eta = \|X^T X - I\|_{\text{op}}$. There is an ENP frame W with $\|X - W\|_F^2 \leq \eta d(d-1)$.*

Proof. If $d = 1$ then $X^T X = na = 1$ and $W = X$ works. Let $d \geq 2$.

Step 1: full spark. Suppose first that every $d \times d$ minor $\mu_S = \det(X_S)^2$ ($S \subset [n]$, $|S| = d$) is nonzero. Consider $F(s) = \frac{1}{2} \log \det(X^T \text{Diag}(e^{2s})X) - a \sum_i s_i$. By Cauchy–Binet,

$$F(s) = \frac{1}{2} \log \sum_{|S|=d} \mu_S e^{T(s)_S}, \quad T(s)_S = 2 \left(\sum_{i \in S} s_i - a \sum_i s_i \right),$$

where $T : \mathbb{R}^n \rightarrow \mathbb{R}^{\binom{n}{d}}$ is linear and $\sum_S T(s)_S = 0$ because each i lies in a fraction $d/n = a$ of the d -sets. The function $\mathcal{E}(y) = \sum_S \mu_S e^{y_S}$ restricted to the subspace $\text{im } T$ has the nonempty sublevel set $\{\mathcal{E} \leq \mathcal{E}(0)\}$, on which $y_S \leq \log(\mathcal{E}(0)/\mu_S)$ for every S and hence, by $\sum_S y_S = 0$, also y_S is bounded below. This set is compact, so $\mathcal{E}|_{\text{im } T}$ attains its minimum at some $T(s^*)$, and s^* minimises F .

Let $D = \text{Diag}(e^{s^*})$, $M = (X^T D^2 X)^{-1/2}$ and $W = DXM$. The computation in the proof of [Lemma 2.4](#) (which does not use $X^T X = I$) gives $\partial_i F = \|w_i\|^2 - a$, so W is ENP, and since $w_i = e^{s_i^*} M x_i$ has norm $\sqrt{a}$,

$$w_i = \sqrt{a} \frac{M x_i}{\|M x_i\|}.$$

Step 2: ordered coordinates. Replacing X, W by XO, WO for a suitable $O \in O(d)$ changes neither the hypotheses nor the distance, and makes $M = \text{Diag}(\lambda_1, \dots, \lambda_d)$ with $0 < \lambda_1 \leq \dots \leq \lambda_d$.

Then $w_{ij} = \lambda_j x_{ij} / \sqrt{c_i}$ with $c_i = a^{-1} \sum_\ell \lambda_\ell^2 x_{i\ell}^2$, so w_{ij} has the sign of x_{ij} and $w_{ij}^2 - x_{ij}^2 = x_{ij}^2 (\lambda_j^2 / c_i - 1)$ is nonpositive and then nonnegative as j increases. As $\sum_j (x_{ij}^2 - w_{ij}^2) = a - a = 0$, the prefix sums $\Delta_i(k) = \sum_{j \leq k} (x_{ij}^2 - w_{ij}^2)$ are nonnegative, vanish at $k = d$, and

$$\sum_j |x_{ij}^2 - w_{ij}^2| = 2 \max_k \Delta_i(k) \leq 2 \sum_{k=1}^{d-1} \Delta_i(k).$$

Since $W^T W = I$ and $(X^T X)_{jj} \leq 1 + \eta$, $\sum_i \Delta_i(k) = \sum_{j \leq k} ((X^T X)_{jj} - 1) \leq k\eta$. Equal signs give $(x_{ij} - w_{ij})^2 \leq |x_{ij}^2 - w_{ij}^2|$, and therefore

$$\|X - W\|_F^2 \leq 2\eta \sum_{k=1}^{d-1} k = \eta d(d-1).$$

Step 3: general X . Let G be a Vandermonde matrix with rows $(1, \theta_i, \dots, \theta_i^{d-1})$ for distinct θ_i ; all its maximal minors are nonzero. Each maximal minor of $X + hG$ is a polynomial in h with nonzero leading coefficient, so for all but finitely many h the matrix $X + hG$ is full spark; choose such $h_k \rightarrow 0$ (so small that no row of $X + h_k G$ vanishes) and normalise each row of $X + h_k G$ to length $\sqrt{a}$ (a positive diagonal left factor, which preserves full spark). The resulting $X_k \rightarrow X$ have ENP frames W_k with $\|X_k - W_k\|_F^2 \leq \eta_k d(d-1)$, $\eta_k \rightarrow \eta$. The set of ENP frames is compact; a limit point W of (W_k) is ENP and satisfies the bound. $\square$

Corollary 3.2. *For all $1 \leq d \leq n$ an ENP frame exists. Consequently every equal-norm $X \in \mathbb{R}^{n \times d}$ is within $\|X - W\|_F^2 \leq 2\|X\|_F^2 + 2\|W\|_F^2 = 4d$ of an ENP frame W .*

Proof. Apply Lemma 3.1 to $x_i = \sqrt{a} e_1$ for all i . $\square$

4 The many-row seed

Theorem 4.1. *There are absolute constants $B, c_*, C_* > 0$ such that the following holds. Let $2 \leq d$, $n \geq Bd^2$, let $X \in \mathbb{R}^{n \times d}$ have $\|x_i\|^2 = a = d/n$ for all i , and suppose $\eta := \|X^T X - I\|_{\text{op}}$ satisfies $0 < \eta \leq c_*$. Then there is an ENP frame W with $\|X - W\|_F^2 \leq C_* \eta d$.*

Throughout, $S = X^T X$, so $\|S - I\|_{\text{op}} \leq \eta \leq c_* \leq \frac{1}{2}$ (the choice of c_* below gives $c_* \leq \frac{1}{64}$), and $e_i = x_i / \sqrt{a}$.

4.1 The noise

In the Frobenius space $\mathbb{R}^{n \times d}$ consider the subspaces

$$\mathcal{E} = \{Z : \langle x_i, z_i \rangle = 0 \ \forall i\}, \quad \mathcal{F} = \{Z \in \mathcal{E} : X^T Z + Z^T X = 0\},$$

and let $m = \dim \mathcal{E} - \dim \mathcal{F} \leq \dim \text{Sym}_d \leq d^2 \leq n$. For a standard Gaussian $g \in \mathbb{R}^{n \times d}$ put

$$Z_0 = n^{-1/2} \Pi_{\mathcal{E}} g, \quad Z = n^{-1/2} \Pi_{\mathcal{F}} g, \quad Z_\perp = Z_0 - Z = n^{-1/2} \Pi_{\mathcal{E} \ominus \mathcal{F}} g.$$

Z and $Z_\perp$ are independent. The rows $z_{0,i} = n^{-1/2} (I - e_i e_i^T) g_i$ of Z_0 are independent. Since $\text{Cov}(Z) \preceq \text{Cov}(Z_0)$, each row z_i of Z is Gaussian with covariance $\Sigma_i \preceq n^{-1} (I - e_i e_i^T)$, and $\text{Cov}(\text{vec } Z) \preceq I/n$. Moreover

$$\mathbb{E} \|Z\|_F^2 = \frac{n(d-1) - m}{n} \leq d, \quad \mathbb{E} \|Z - Z_0\|_F^2 = \frac{m}{n} \leq \frac{d^2}{n}. \quad (4.1)$$

Lemma 4.2 (Row moments). *Let $r_i = \|z_i\|^2/a$, $\mu_i = \mathbb{E}r_i$ and $\delta_i = 1 - \mu_i$, and suppose $d^2 \leq n$ (as is the case under the hypotheses of [Theorem 4.1](#)). Then $1/d \leq \delta_i \leq 1$, $\sum_i a\delta_i = 1 + m/n \leq 2$, and*

$$\mathcal{M} := a \sum_i \mathbb{E}[(r_i - 1)^2(1 + r_i)^2] \leq C_{\mathcal{M}},$$

with $C_{\mathcal{M}}$ absolute.

Proof. $\tilde{\Sigma}_i = \Sigma_i/a \preceq d^{-1}I$ has trace $\mu_i \leq (d-1)/d$, so $\text{tr } \tilde{\Sigma}_i^2 \leq 1/d$ and $\text{tr } \tilde{\Sigma}_i^4 \leq d^{-3}$. For $\xi_i = r_i - \mu_i$ the Gaussian moment formulas give $\mathbb{E}\xi_i^2 = 2 \text{tr } \tilde{\Sigma}_i^2 \leq 2/d$ and $\mathbb{E}\xi_i^4 = 12(\text{tr } \tilde{\Sigma}_i^2)^2 + 48 \text{tr } \tilde{\Sigma}_i^4 \leq 60/d^2$. Next $\sum_i a\delta_i = d - \mathbb{E}\|Z\|_F^2 = 1 + m/n$ by (4.1). Finally $(r_i - 1)^2 \leq 2\xi_i^2 + 2\delta_i^2$ and $(1 + r_i)^2 \leq 8 + 2\xi_i^2$, so

$$\mathbb{E}(r_i - 1)^2(1 + r_i)^2 \leq 16\mathbb{E}\xi_i^2 + 4\mathbb{E}\xi_i^4 + 16\delta_i^2 + 4\delta_i^2\mathbb{E}\xi_i^2 \leq \frac{32}{d} + \frac{240}{d^2} + \left(16 + \frac{8}{d}\right)\delta_i^2,$$

and, as $\delta_i^2 \leq \delta_i$ and $na = d$, $\mathcal{M} \leq 32 + \frac{240}{d} + (16 + \frac{8}{d})a \sum_i \delta_i \leq 32 + 120 + 2 \cdot 20 = 192$ for $d \geq 2$; so $C_{\mathcal{M}} = 192$ works. $\square$

4.2 Row normalisation and the exact second-order identity

For $0 < t \leq 1$ set

$$v_i = \frac{x_i + tz_i}{\sqrt{1 + t^2 r_i}}, \quad v_{0,i} = \frac{x_i + tz_{0,i}}{\sqrt{1 + t^2 r_{0,i}}}, \quad r_{0,i} = \|z_{0,i}\|^2/a.$$

Since $z_i \perp x_i$, $\|v_i\|^2 = \|v_{0,i}\|^2 = a$ exactly. The map $y \mapsto \sqrt{a}y/\|y\|$ is the metric projection onto the closed ball of radius $\sqrt{a}$ on $\{\|y\| \geq \sqrt{a}\}$, hence 1-Lipschitz there. Thus

$$\|V - X\|_F \leq t\|Z\|_F, \quad \|V - V_0\|_F \leq t\|Z - Z_0\|_F, \quad \|V\|_{\text{op}} \leq \|X + tZ\|_{\text{op}}, \quad \|V_0\|_{\text{op}} \leq \|X + tZ_0\|_{\text{op}}, \quad (4.2)$$

the last two because $V = \text{Diag}(\lambda)(X + tZ)$ with $0 < \lambda_i \leq 1$, and similarly for V_0 .

Expanding $1/(1 + t^2 r_i) = 1 - t^2 c_i$ with

$$c_i = \frac{r_i}{1 + t^2 r_i},$$

and using $X^T Z + Z^T X = 0$, one finds the exact identity

$$V^T V = S + t^2 Q(Z) + R_3 + R_4, \quad Q(Z) = Z^T Z - \sum_i r_i x_i x_i^T, \quad (4.3)$$

$$R_3 = -t^3 \sum_i c_i (x_i z_i^T + z_i x_i^T), \quad R_4 = -t^4 \sum_i c_i (z_i z_i^T - r_i x_i x_i^T).$$

Lemma 4.3 (Mean and fluctuation of Q). $\|\mathbb{E}Q(Z) - (I - S)\|_{\text{op}} \leq m/n$ and $\mathbb{E}\|Q(Z) - \mathbb{E}Q(Z)\|_F^2 \leq 8d^2/n$.

Proof. For Z_0 : $\mathbb{E}Z_0^T Z_0 = n^{-1} \sum_i (I - e_i e_i^T) = I - S/d$ and $\mathbb{E}r_{0,i} = (d-1)/d$, so $\mathbb{E}Q(Z_0) = I - S/d - (1 - 1/d)S = I - S$. As $Z_0 = Z + Z_{\perp}$ with independent centred summands and Q is quadratic, $\mathbb{E}Q(Z) = \mathbb{E}Q(Z_0) - \mathbb{E}Q(Z_{\perp})$, and $\mathbb{E}Q(Z_{\perp})$ is the difference of the positive semidefinite matrices $\mathbb{E}Z_{\perp}^T Z_{\perp}$ and $\sum_i \mathbb{E}(\|z_{\perp,i}\|^2/a)x_i x_i^T$, both of trace $\mathbb{E}\|Z_{\perp}\|_F^2 = m/n$; so its norm is at most m/n .

Entry (α, β) of Q is the quadratic form $\sum_i z_i^T \mathbf{B}_{\alpha\beta,i} z_i$ with $\mathbf{B}_{\alpha\beta,i} = \frac{1}{2}(E_{\alpha\beta} + E_{\beta\alpha}) - (e_i)_\alpha (e_i)_\beta I$, where the $E_{\alpha\beta}$ are the matrix units of $\mathbb{R}^{d \times d}$. For a Gaussian vector with covariance $\Sigma \preceq I/n$ and symmetric coefficient matrix $\mathbf{B}$, the variance of the quadratic form is $2\|\Sigma^{1/2} \mathbf{B} \Sigma^{1/2}\|_F^2 \leq 2\|\mathbf{B}\|_F^2/n^2$; we apply this to $\text{vec } Z$ and the block-diagonal matrix with blocks $\mathbf{B}_{\alpha\beta,i}$. Since $\sum_{\alpha,\beta} \|\mathbf{B}_{\alpha\beta,i}\|_F^2 \leq 2d^2 + 2d \leq 4d^2$ for each i , summing over α, β and the n blocks gives the claim. $\square$

Lemma 4.4 (Remainder). *Let $\bar{c} = 1/(1+t^2)$, $\Gamma = \text{Diag}(c_i - \bar{c})$ and $\Lambda = \text{Diag}(c_i r_i - \bar{c})$. Then*

$$\|R_3\|_{\text{op}} \leq 2t^3 \|X\|_{\text{op}} \|\Gamma Z\|_F, \quad \|R_4\|_{\text{op}} \leq t^4 (\|Z\|_{\text{op}}^2 + \|Z\|_{\text{op}} \|\Gamma Z\|_F + \|S\|_{\text{op}} + \|X\|_{\text{op}} \|\Lambda X\|_F),$$

and $\mathbb{E}\|\Gamma Z\|_F^2 \leq \mathcal{M}$, $\mathbb{E}\|\Lambda X\|_F^2 \leq \mathcal{M}$.

Proof. Because $\sum_i (x_i z_i^T + z_i x_i^T) = X^T Z + Z^T X = 0$, the constant part $\bar{c}$ of the weights drops out of R_3 : $R_3 = -t^3(X^T \Gamma Z + Z^T \Gamma X)$, which gives the first bound. Next $\sum_i c_i z_i z_i^T = \bar{c} Z^T Z + Z^T \Gamma Z$ and $\sum_i c_i r_i x_i x_i^T = \bar{c} S + X^T \Lambda X$, with $\|Z^T \Gamma Z\|_{\text{op}} \leq \|Z\|_{\text{op}} \|\Gamma Z\|_F$ and similarly for Λ , and $\bar{c} \leq 1$. For the moments,

$$c_i - \bar{c} = \frac{r_i - 1}{(1+t^2 r_i)(1+t^2)}, \quad c_i r_i - \bar{c} = \frac{(r_i - 1)(1+r_i+t^2 r_i)}{(1+t^2 r_i)(1+t^2)},$$

so $|c_i - \bar{c}| \leq |r_i - 1|$ and, as $1+r+t^2 r \leq (1+r)(1+t^2 r)$, $|c_i r_i - \bar{c}| \leq |r_i - 1|(1+r_i)$. Hence $\|\Gamma Z\|_F^2 = a \sum_i (c_i - \bar{c})^2 r_i \leq a \sum_i (r_i - 1)^2 (1+r_i)^2$ and $\|\Lambda X\|_F^2 = a \sum_i (c_i r_i - \bar{c})^2 \leq a \sum_i (r_i - 1)^2 (1+r_i)^2$; take expectations and use Lemma 4.2. $\square$

Remark 4.5. The termwise bound $\|R_3\|_{\text{op}} \leq 2t^3 \sum_i c_i \|x_i\| \|z_i\| \approx 2t^3 d$ would only allow $\eta \lesssim d^{-2}$. Centring the weights at $\bar{c}$, which is allowed by the first-order constraint, replaces the n terms of size a by fluctuations of relative size $d^{-1/2}$, whose ℓ_2 -sum over the rows is $O(1)$.

4.3 A dense reference graph

Lemma 4.6. *There are absolute $h, t_0 \in (0, 1]$ and $c_1 > 0$ such that for $d \geq 2$ and $t \leq t_0$, with probability at least $1 - ne^{-c_1(n-1)}$, every i has at least $0.9(n-1)$ indices $j \neq i$ with*

$$|\langle v_{0,i}, v_{0,j} \rangle| \geq ht\sqrt{a/n}.$$

Proof. Write $z_{0,j}/\sqrt{a} = d^{-1/2} g'_j$ with $g'_j \sim N(0, I - e_j e_j^T)$ independent. Fix i and condition on row i ; let $f = v_{0,i}/\sqrt{a}$, a unit vector. For $j \neq i$,

$$\frac{\langle v_{0,i}, v_{0,j} \rangle}{a} = \frac{\nu_j}{N_j}, \quad \nu_j = u_j + \frac{t}{\sqrt{d}} \langle f, g'_j \rangle, \quad u_j = \langle f, e_j \rangle, \quad N_j = (1 + \frac{t^2}{d} \|g'_j\|^2)^{1/2} \geq 1,$$

and the target inequality reads $|\nu_j|/N_j \geq ht/\sqrt{d}$. By Markov, $\mathbb{P}(N_j > 2) \leq t^2/3$. On $\{N_j \leq 2\}$ failure requires $|\nu_j| < 2ht/\sqrt{d}$. Here $\nu_j \sim N(u_j, t^2(1 - u_j^2)/d)$. If $|u_j| \leq \frac{1}{2}$ the variance is at least $3t^2/(4d)$ and the Gaussian density bound gives probability at most $4h/\sqrt{3\pi/2} \leq 3h$. If $|u_j| > \frac{1}{2}$ and $2ht \leq \frac{1}{4}$, failure requires $\frac{t}{\sqrt{d}} |\langle f, g'_j \rangle| > \frac{1}{4}$, of probability at most $16t^2/d$. Choose h and t_0 so that $3h + t_0^2/3 + 8t_0^2 \leq \frac{1}{50}$, for instance $h = \frac{1}{300}$ and $t_0 = \frac{1}{30}$ (this also gives $2ht \leq \frac{1}{4}$, and $16t^2/d \leq 8t^2$ as $d \geq 2$). The failure indicators for $j \neq i$ are conditionally independent with probabilities at most $\frac{1}{50}$, so by the exponential Markov inequality (with parameter 1) more than $0.1(n-1)$ failures have probability at most $\exp(-(n-1)[0.1 - \frac{e-1}{50}]) \leq e^{-c_1(n-1)}$. Take a union bound over i . $\square$

4.4 Proof of Theorem 4.1

Constants. Fix h, t_0, c_1 from Lemma 4.6. Put $\kappa_0 = h^2/200$, $C' = 36^2 \cdot 20$, and $C_{\text{mix}} = 125/h^2 + 3$, then $\Theta = \max\{42, C_{\text{mix}}\}$ and $\theta = \min\{1/(4\Theta), h/60\}$. Let $C_R = 300 + 100\sqrt{C_{\mathcal{M}}}$, choose $t_1 = \min\{t_0, \theta/(16C_R)\}$, and set

$$t^2 = 4\eta/\theta, \quad c_* = \theta t_1^2/4,$$

so that $\eta \leq c_*$ gives $t \leq t_1$. Finally choose B so large that $n \geq Bd^2$ implies $d^2/n \leq \theta/4$, $\sqrt{160d^2/n} \leq \theta/4$, $2aC'd/\kappa_0 \leq \frac{1}{4}$ (this bounds $\sum_{i \in \mathcal{B}_0} 2a$ for the set $\mathcal{B}_0$ below, which has $|\mathcal{B}_0| \leq C'd/\kappa_0$), $n \geq 30$, and $ne^{-c_1(n-1)} \leq 0.01$.

The event. By Markov's inequality, (4.1), Lemmas 4.3 and 4.4, the net bound (Lemma A.1(c)) for Z and Z_0 , and Lemma 4.6, with positive probability all of the following hold:

$$\|Z\|_{\text{F}}^2 \leq 20d, \quad \|Z - Z_0\|_{\text{F}}^2 \leq \frac{20d^2}{n}, \quad \|Q - \mathbb{E}Q\|_{\text{F}}^2 \leq \frac{160d^2}{n}, \quad \|\Gamma Z\|_{\text{F}}^2 \leq 20\mathcal{M}, \quad \|\Lambda X\|_{\text{F}}^2 \leq 20\mathcal{M},$$

$\|Z\|_{\text{op}}, \|Z_0\|_{\text{op}} \leq 16$, and the conclusion of Lemma 4.6: the five Markov events fail with probability at most $\frac{1}{20}$ each, the two net bounds (Lemma A.1(c) with $\sigma^2 = 1/n$, $u = 16$) with probability at most $2 \cdot 5^{n+d}e^{-8n}$ each, and Lemma 4.6 with probability at most 0.01. Fix such a sample.

Spectral error. By (4.3), $V^T V - I = (1 - t^2)(S - I) + t^2(\mathbb{E}Q - (I - S)) + t^2(Q - \mathbb{E}Q) + R_3 + R_4$. Since $\|X\|_{\text{op}} \leq \sqrt{2}$, $\|S\|_{\text{op}} \leq \frac{3}{2}$ and $t \leq 1$, Lemma 4.4 and the event give $\|R_3\|_{\text{op}} + \|R_4\|_{\text{op}} \leq t^3(2\sqrt{40\mathcal{M}} + 256 + 16\sqrt{20\mathcal{M}} + \frac{3}{2} + \sqrt{40\mathcal{M}}) \leq C_R t^3$, and therefore

$$\delta := \|V^T V - I\|_{\text{op}} \leq \eta + t^2 \frac{d^2}{n} + t^2 \sqrt{\frac{160d^2}{n}} + C_R t^3 \leq \frac{\theta}{4} t^2 + \frac{\theta}{4} t^2 + \frac{\theta}{4} t^2 + \frac{\theta}{16} t^2 < \theta t^2.$$

Graph. By (4.2), $\|V\|_{\text{op}}, \|V_0\|_{\text{op}} \leq \sqrt{2} + 16 \leq 18$ and $\|VV^T - V_0V_0^T\|_{\text{F}} \leq 36t\|Z - Z_0\|_{\text{F}}$, so $\|VV^T - V_0V_0^T\|_{\text{F}}^2 \leq C't^2d^2/n$. Let $\mathcal{B}_0$ be the set of rows whose contribution to this sum exceeds $\kappa_0 at^2$; then $|\mathcal{B}_0| \leq C'd/\kappa_0$. A row $i \notin \mathcal{B}_0$ has at most $\kappa_0 at^2/((h/2)^2 t^2 a/n) = n/50$ indices with $|(VV^T - V_0V_0^T)_{ij}| \geq \frac{h}{2} t \sqrt{a/n}$, so by Lemma 4.6 it has at least $0.85n$ indices $j \neq i$ with $|(VV^T)_{ij}| \geq \frac{h}{2} t \sqrt{a/n}$.

Let $U = V(V^T V)^{-1/2}$, $P = UU^T$. By Lemma 2.1 (rows of $P - VV^T$) a row loses at most $6a\theta^2 t^4/((h/4)^2 t^2 a/n) \leq n/20$ of these indices, so each $i \notin \mathcal{B}_0$ has at least $0.8n$ indices $j \neq i$ with $P_{ij}^2 \geq (h^2/16)at^2/n$. Also $a/2 \leq P_{ii} \leq 2a$ and $\sum_{i \in \mathcal{B}_0} P_{ii} \leq \frac{1}{4}$. Lemma 2.10 with $\mathcal{B} = \mathcal{B}_0$, part (i) with $\vartheta = 0.3$ and $\gamma = h^2 at^2/16$, and part (iii) with $\alpha = a$ (so $R = \frac{8}{3a}$, $F_{\mathcal{B}} \leq \frac{4}{3}$) shows that $H(LP) \leq \frac{7/3}{0.3\gamma} + \frac{8}{3a} \leq C_{\text{mix}}/(at^2)$.

Seed. $\|U - X\|_{\text{F}}^2 \leq 2d\delta^2 + 2t^2\|Z\|_{\text{F}}^2 \leq 42t^2d$, the diagonal error is at most $2a\delta \leq 2\theta at^2 \leq at^2/(2\Theta)$, and the barrier constant is at most $\Theta/(at^2)$. So U is a (Θ, t) -seed, and Corollary 2.13 gives an ENP frame within $3\Theta t^2 d = (12\Theta/\theta)\eta d$ of X ; thus $C_* = 12\Theta/\theta$ works. $\square$

5 The moderate-row seed

Theorem 5.1. *There are absolute constants $A, C_m > 0$ and $\varepsilon_0 \in (0, \frac{1}{2}]$ such that the following holds. Let $U \in \mathbb{R}^{n \times d}$ be Parseval with $2d \leq n$, $d \geq A \log(2n)$, and $|||u_i||^2 - a| \leq \varepsilon a$ for all i , where $0 < \varepsilon \leq \varepsilon_0$. Then there is an ENP frame $\hat{U}$ with $\|\hat{U} - U\|_{\text{F}}^2 \leq C_m \varepsilon d$.*

Throughout this section $P = UU^T$, $p = \text{diag } P$, $D_p = \text{Diag}(p)$ and $L = LP$; as $\varepsilon \leq \varepsilon_0 \leq \frac{1}{2}$, $\frac{a}{2} \leq p_i \leq \frac{3a}{2} \leq \frac{3}{4}$. For a symmetric $M \in \mathbb{R}^{n \times n}$, $\mathcal{L}(M) = \sum_{i < j} M_{ij}^2 (e_i - e_j)(e_i - e_j)^T$ is the

Laplacian with weights M_{ij}^2 ; thus $L = \mathcal{L}(P)$, $x^T \mathcal{L}(M)x = \frac{1}{2} \|[\text{Diag } x, M]\|_F^2$, and

$$\mathcal{L}(M) \preceq 2 \max_i \|M_i\|^2 I, \quad \mathcal{L}(M + M') \succeq \frac{1}{2} \mathcal{L}(M) - \mathcal{L}(M'), \quad (5.1)$$

by $(x_i - x_j)^2 \leq 2x_i^2 + 2x_j^2$ and $(v + w)^2 \geq \frac{1}{2}v^2 - w^2$. We write $K = nI - \mathbf{1}\mathbf{1}^T$ for the Laplacian of the complete graph.

Scales. Put $\sigma = at^2$, where $t \asymp \sqrt{\varepsilon}$ is the size of the perturbation. The perturbation creates edges of weight $\gtrsim \sigma/n$, the resulting barrier constant is $O(1/\sigma)$, and so all diagonal errors must be at most a small multiple of σ . The table lists the errors and how each is controlled ($\rho = D^2 t^2$ is the filter level, D a large constant).

diagonal error	size/ σ	reason
input	ε/t^2	$t^2 = K_0 \varepsilon$, K_0 large
linear term $2tAZ$	$6D\sqrt{\log(2n)/d}$	filter (Lemma 5.3), $d \geq A \log 2n$
quadratic fluctuation	η'	concentration, $d \geq A \log 2n$
filter bias $t^2 b_*$	0	cancelled by the drift (Lemmas 5.6 and 5.10)
drift cross terms	$40/D + 1/(2D^2)$	D large
base bias $t^2 b_0$	$12\varepsilon/d$	Lemma 5.6
retraction remainder	$C_E t$	t small

5.1 The horizontal tangent space and the filtered covariance

Fix an isometry $V \in \mathbb{R}^{n \times (n-d)}$ onto $(\text{im } U)^\perp$, and let $\mathcal{Z} = \mathbb{R}^{(n-d) \times d}$ with the Frobenius inner product. For $Z \in \mathcal{Z}$ the matrix VZ is horizontal ($U^T VZ = 0$), and we put

$$Y_Z = VZU^T + UZ^T V^T, \quad \|(Y_Z)_i\|^2 = \|Z^T v_i\|^2 + \|Z u_i\|^2 \leq \|Z\|_{\text{op}}^2, \quad \|Y_Z\|_F^2 = 2\|Z\|_F^2.$$

Define

$$\mathcal{A} : \mathcal{Z} \rightarrow \mathbb{R}^n, \quad (\mathcal{A}Z)_i = v_i^T Z u_i = (VZU^T)_{ii}, \quad \mathcal{A}^* x = N_x := V^T \text{Diag}(x)U.$$

Then $\mathcal{A}\mathcal{A}^* = L$, i.e. $\|N_x\|_F^2 = x^T Lx$ (Lemma 2.2). Let

$$\bar{\mathcal{A}} = D_p^{-1/2} \mathcal{A}, \quad S = \bar{\mathcal{A}} \bar{\mathcal{A}}^* = D_p^{-1/2} L D_p^{-1/2}, \quad \Omega = \bar{\mathcal{A}}^* \bar{\mathcal{A}}.$$

Since $L \succeq 0$ and $S = I - D_p^{-1/2} (P \circ P) D_p^{-1/2}$ with $P \circ P \succeq 0$ (Schur product theorem),

$$0 \preceq S \preceq I, \quad 0 \preceq \Omega \preceq I, \quad \text{tr}(I - S) = \sum_i \frac{P_{ii}^2}{p_i} = d. \quad (5.2)$$

Fix an orthonormal eigenbasis of S , and for each of its members y with eigenvalue $\mu > 0$ put $f_y = D_p^{-1/2} y$ and $Z_y = \mu^{-1/2} \bar{\mathcal{A}}^* y = \mu^{-1/2} N_{f_y}$. Since $\Omega Z_y = \mu Z_y$ and $\langle Z_y, Z_{y'} \rangle = \langle y, Sy' \rangle / \sqrt{\mu \mu'}$, the Z_y form an orthonormal eigenbasis of Ω on $(\ker \Omega)^\perp$. Sums $\sum_{\mu > 0}$ below run over these basis members (with multiplicity). Since $p_i \geq a/2$,

$$\|f_y\|_\infty \leq \sqrt{2/a}. \quad (5.3)$$

Definition 5.2 (Filtered covariance). For $\rho > 0$ let $C_\rho = \rho(I - \Omega)(\rho I + \Omega)^{-1}$ on $\mathcal{Z}$, let $Z \sim N(0, C_\rho/n)$, and $Y = Y_Z$.

Lemma 5.3. (a) $0 \preceq C_\rho \preceq I$, and $C_\rho = I$ on $\ker \Omega$.

(b) $\text{Cov}(\mathcal{A}Z) = n^{-1}D_p^{1/2} \rho S(I - S)(\rho I + S)^{-1}D_p^{1/2} \preceq (\rho/n)D_p$.

(c) $I - C_\rho = \Omega + R_\rho$ with $R_\rho = \sum_{\mu>0} r_\mu Z_y \otimes Z_y$, $r_\mu = \mu(1 - \mu)/(\rho + \mu) \geq 0$, and

$$\sum_{\mu>0} r_\mu \leq d, \quad \sum_{\mu>0} \frac{r_\mu}{\sqrt{\mu}} \leq \frac{d}{2\sqrt{\rho}}, \quad \sum_{\mu>0} \frac{r_\mu}{\mu} \leq \frac{d}{\rho}.$$

Proof. (a) The eigenvalues of C_ρ are $\rho(1 - \mu)/(\rho + \mu) \in [0, 1]$. (b) By singular value calculus $\mathcal{A}C_\rho\mathcal{A}^* = \rho S(I - S)(\rho I + S)^{-1}$, whose eigenvalues $\rho\mu(1 - \mu)/(\rho + \mu)$ are at most ρ . (c) On the μ -eigenspace, $1 - \rho(1 - \mu)/(\rho + \mu) = \mu + \mu(1 - \mu)/(\rho + \mu)$. The sums follow from $\mu/(\rho + \mu) \leq 1$, $\sqrt{\mu}/(\rho + \mu) \leq 1/(2\sqrt{\rho})$, $1/(\rho + \mu) \leq 1/\rho$, and $\sum_{\mu>0}(1 - \mu) \leq \text{tr}(I - S) = d$. $\square$

Remark 5.4. Only these properties of the filter $\phi(\mu) = \rho(1 - \mu)/(\rho + \mu)$ are used: $\phi(0) = 1$ (so $C_\rho = I$ on $\ker \Omega$), $0 \leq \phi(\mu) \leq 1 - \mu$, $\mu\phi(\mu) \leq \rho$, and the sums in (c). The hard cutoff $(1 - \mu)\mathbf{1}[\mu < \rho]$ satisfies them with $\sum r_\mu/\sqrt{\mu} \leq d/\sqrt{\rho}$, which doubles some constants below; the rational filter keeps every object a rational function of S , which is convenient for formalisation.

5.2 Commutator identities

For $Z \in \mathcal{Z}$ define the quadratic diagonal and its polarisation

$$q_i(Z) = \|Z^T v_i\|^2 - \|Z u_i\|^2, \quad q_i(Z, M) = \langle Z^T v_i, M^T v_i \rangle - \langle Z u_i, M u_i \rangle,$$

so that $q(Z + M) = q(Z) + 2q(Z, M) + q(M)$ and, by Cauchy–Schwarz,

$$|q_i(Z, M)| \leq \|(Y_Z)_i\| \|(Y_M)_i\|, \quad |q_i(M)| \leq \|(Y_M)_i\|^2. \quad (5.4)$$

Lemma 5.5. Let $f \in \mathbb{R}^n$, $F = \text{Diag}(f)$, and $R = I - 2P$ (an orthogonal matrix). Then:

(a) $q(N_f) = \mathcal{A}\Gamma_f$ with $\Gamma_f = V^T F R F U$, and $\|\Gamma_f\|_F \leq 2\|f\|_\infty \|N_f\|_F$;

(b) $T_f := Y_{N_f} = FP + PF - 2PFP$ satisfies $[X, T_f] = RF[X, P] + [X, P]FR$ for diagonal X ; hence $\mathcal{L}(T_f) \preceq 4\|f\|_\infty^2 L$, and $x^T \mathcal{L}(T_{e_k})x \leq 2 \sum_j P_{kj}^2 (x_k - x_j)^2$.

Proof. (a) For $x \in \mathbb{R}^n$ put $X = \text{Diag}(x)$, $B_x = V^T X V$, $A_x = U^T X U$. Then $\langle x, q(Z) \rangle = \langle Z, B_x Z - Z A_x \rangle$ for every Z , and, since X and F commute,

$$B_x N_f - N_f A_x = V^T [X(I - P)F - F P X] U = V^T [F(I - P)X - X P F] U = B_f N_x - N_x A_f.$$

Hence $\langle x, q(N_f) \rangle = \langle N_f, B_f N_x - N_x A_f \rangle = \langle B_f N_f - N_f A_f, N_x \rangle$. As $\langle M, N_x \rangle = \sum_i x_i v_i^T M u_i = \langle \mathcal{A}M, x \rangle$ for every M , this says $q(N_f) = \mathcal{A}(B_f N_f - N_f A_f)$, and $B_f N_f - N_f A_f = V^T F(I - P)F U - V^T F P F U = \Gamma_f$. The norm bound follows from $\|B_f\|_{\text{op}}, \|A_f\|_{\text{op}} \leq \|f\|_\infty$.

(b) $Y_{N_f} = V V^T F U U^T + U U^T F V V^T = (I - P)FP + PF(I - P)$. Since X commutes with F , $[X, FP] = F[X, P]$, $[X, PF] = [X, P]F$ and $[X, PFP] = [X, P]FP + PF[X, P]$, which gives the commutator formula. Then $\|[X, T_f]\|_F \leq 2\|f\|_\infty \|[X, P]\|_F$, i.e. the first bound (Lemma 2.2). For $f = e_k$ and $E_k = \text{Diag}(e_k)$, $E_k[X, P]$ and $[X, P]E_k$ are the k th row and column of $[X, P]$, each of squared norm $\sum_j P_{kj}^2 (x_k - x_j)^2$. $\square$

5.3 The second-order mean and the drift

Lemma 5.6 (Mean). $\mathbb{E}q(Z) = a\mathbf{1} - p - b_0 - b_*$ with $b_* := \mathcal{A}m_*$, where

$$b_0 = \frac{1}{n}L(p^{-1}), \quad \|b_0\|_\infty \leq \frac{12\varepsilon}{n}, \quad m_* = \frac{1}{n} \sum_{\mu>0} \frac{r_\mu}{\mu} \Gamma_{f_y}, \quad \|m_*\|_F^2 \leq \frac{2a}{\rho}.$$

Proof. For $G \sim N(0, I)$ on $\mathcal{Z}$, $\mathbb{E}\|G^T v_i\|^2 = d(1 - p_i)$ and $\mathbb{E}\|Gu_i\|^2 = (n - d)p_i$, so the unfiltered covariance I/n contributes $a\mathbf{1} - p$. The covariance $\Omega = \sum_j \zeta_j \otimes \zeta_j$, $\zeta_j = \bar{\mathcal{A}}^* e_j = v_j u_j^T / \sqrt{p_j}$, contributes

$$\frac{1}{n} \sum_j q_i(\zeta_j) = \frac{1}{n} \sum_j \frac{(I - P)_{ij}^2 p_j - (1 - p_j) P_{ij}^2}{p_j} = \frac{1}{n} \left(1 - \sum_j \frac{P_{ij}^2}{p_j} \right) = \frac{1}{n} (Lp^{-1})_i,$$

using $\sum_j (I - P)_{ij}^2 = 1 - p_i$ and $\sum_j P_{ij}^2 = p_i$. Here $(Lp^{-1})_i = \sum_j P_{ij}^2 (p_i^{-1} - p_j^{-1})$ and $|p_i^{-1} - p_j^{-1}| \leq 2\varepsilon a / ((1 - \varepsilon)a)^2 \leq 8\varepsilon/a$, so $|(Lp^{-1})_i| \leq 12\varepsilon$. Finally R_ρ/n contributes $\frac{1}{n} \sum_\mu r_\mu q(Z_y) = \frac{1}{n} \sum_\mu \frac{r_\mu}{\mu} q(N_{f_y}) = \mathcal{A}m_*$ by Lemma 5.5(a), and with $\|N_{f_y}\|_F = \sqrt{\mu}$, (5.3) and Lemma 5.3(c),

$$\|m_*\|_F \leq \frac{2}{n} \sqrt{\frac{2}{a}} \sum_{\mu>0} \frac{r_\mu}{\sqrt{\mu}} \leq \frac{2}{n} \sqrt{\frac{2}{a}} \cdot \frac{d}{2\sqrt{\rho}} = \sqrt{\frac{2a}{\rho}}. \quad \square$$

Remark 5.7 (Why a drift). The filter removes covariance from the directions Z_y , and with it their share $t^2 b_* = t^2 \mathcal{A}m_*$ of the second-order drift towards $a\mathbf{1}$. This bias need not be below the tolerance $\sigma/(12\Theta)$ used later: for instance, for many small blocks joined through a common row it is of order at^2 at that row. (Since $\Gamma_f = V^T[(ff^T) \circ (I - 2P)]U$ is quadratic in f , and kernel modes $Sy = 0$ have $N_{f_y} = 0$ and hence $\Gamma_{f_y} = 0$, the spectral identity $\sum_y \frac{1-\mu}{\rho+\mu} yy^T = (I - S)(\rho I + S)^{-1}$ shows that m_* does not depend on the choice of eigenbasis: $m_* = \frac{1}{n} V^T [(D_p^{-1/2}(I - S)(\rho I + S)^{-1} D_p^{-1/2}) \circ (I - 2P)]U$.) However, $t^2 b_*$ is exactly the *first-order* diagonal change produced by the deterministic horizontal direction $\frac{t}{2} m_*$, whose norm is $\frac{t}{2} \|m_*\|_F \leq \sqrt{a}/2/D$. Adding this direction to the noise cancels the bias exactly, at the price of cross terms of size $O(\sigma/D)$.

5.4 The expected graph

Since $\mathbb{E}Y_{ij}^2$ is linear in the covariance, $\mathbb{E}_\Sigma \mathcal{L}(Y) = \sum_k \lambda_k \mathcal{L}(Y_{\psi_k})$ for every representation $\Sigma = \sum_k \lambda_k \psi_k \otimes \psi_k$ with real λ_k ; we use it for $C_\rho/n = (I - \Omega - R_\rho)/n$ with $\Omega = \sum_j \zeta_j \otimes \zeta_j$.

Lemma 5.8. $\mathbb{E}\mathcal{L}(Y) \succeq \frac{a}{4n} \mathbf{K} - \left(\frac{2}{n} + \frac{8}{d} + \frac{8}{\rho} \right) L$.

Proof. Unfiltered noise. For $Z_0 \sim N(0, I/n)$ and $i \neq j$, $(Y_{Z_0})_{ij} = v_i^T Z_0 u_j + v_j^T Z_0 u_i$ has variance $n^{-1} \|v_i u_j^T + v_j u_i^T\|_F^2 = n^{-1} (p_i + p_j - 2p_i p_j - 2P_{ij}^2)$. As $p \leq \frac{3}{4}$, $p_i + p_j - 2p_i p_j \geq \frac{1}{4} (p_i + p_j) \geq \frac{a}{4}$, so $\mathbb{E}\mathcal{L}(Y_{Z_0}) \succeq \frac{a}{4n} \mathbf{K} - \frac{2}{n} L$.

Base loss. $Y_{\zeta_k} = T_{e_k} / \sqrt{p_k}$, so by Lemma 5.5(b) and $p_k \geq a/2$, $\sum_k x^T \mathcal{L}(Y_{\zeta_k}) x \leq \frac{2}{a} \cdot 2 \sum_{k,j} P_{kj}^2 (x_k - x_j)^2 = \frac{8}{a} x^T L x$; the base part Ω/n therefore removes at most $\frac{8}{d} L$.

Residual loss. $\mathcal{L}(Y_{Z_y}) = \mu^{-1} \mathcal{L}(T_{f_y}) \preceq \frac{8}{a\mu} L$ by Lemma 5.5(b) and (5.3), so R_ρ/n removes at most $\frac{8}{an} \sum_\mu \frac{r_\mu}{\mu} L \leq \frac{8}{\rho} L$. $\square$

5.5 A good sample

Let $\ell_i = \frac{1}{n} \sum_{\mu} r_{\mu} \|(Y_{Z_y})_i\|^2$ be the variance that R_{ρ}/n removes from row i . Since $\|Y_{Z_y}\|_{\mathbb{F}}^2 = 2$, $\sum_i \ell_i \leq \frac{2}{n} \sum_{\mu} r_{\mu} \leq 2a$. Fix $\delta_0 = 1/6400$ and let

$$\mathfrak{B} = \{i : \ell_i > \delta_0 a\}, \quad b = |\mathfrak{B}| \leq 2/\delta_0 =: b_{\max},$$

a deterministic set depending only on U and ρ . Write $\mathcal{C}(Y) = \sum_{i < j} 2P_{ij}Y_{ij}(e_i - e_j)(e_i - e_j)^T$, so that $\mathcal{L}(P + tY) = L + t\mathcal{C}(Y) + t^2\mathcal{L}(Y)$.

Lemma 5.9. *There is an absolute $h \in (0, 1)$ and, for each $\eta' \in (0, \frac{1}{32}]$, a constant $A_{\eta'}$ such that the following holds for all $\rho > 0$ and $0 < t \leq 1$ whenever $d \geq A_{\eta'} \log(2n)$. With positive probability:*

- (S1) $\|Z\|_{\text{op}} \leq 16$, $\max_i \|Z^T v_i\|^2 \leq 3a$ and $\max_i \|Y_i\|^2 \leq 800a$;
- (S2) $\max_i |(\mathcal{A}Z)_i| \leq 3\sqrt{\rho a \log(2n)/n}$ and $\max_i |q_i(Z) - \mathbb{E}q_i(Z)| \leq \eta'a$;
- (S3) every $i \notin \mathfrak{B}$ has at least $0.95n$ indices $j \neq i$ with $|P_{ij} + tY_{ij}| \geq ht\sqrt{a/n}$;
- (S4) for every x supported on $\mathfrak{B}$,

$$|tx^T \mathcal{C}(Y)x| \leq \eta'(x^T Lx + at^2\|x\|^2), \quad |x^T (\mathcal{L}(Y) - \mathbb{E}\mathcal{L}(Y))x| \leq \eta'a\|x\|^2.$$

Proof. Since $C_{\rho} \preceq I$, every linear functional $\langle Z, \Psi \rangle$ has variance at most $\|\Psi\|_{\mathbb{F}}^2/n$; we use the tools of [Appendix A](#).

(S1): The net bound ([Lemma A.1\(c\)](#)) with $\sigma^2 = 1/n$ gives $\mathbb{P}(\|Z\|_{\text{op}} > 16) \leq 2 \cdot 5^n e^{-8n}$. The vector $Z^T v_i$ has covariance $\preceq n^{-1}I_d$ and trace $\leq a$; [Lemma A.1\(a\)](#) with $u = 2a$ (so $\min\{u^2/\|\cdot\|_{\mathbb{F}}^2, u/\|\cdot\|_{\text{op}}\} \geq 2d$) and a union bound give $\max_i \|Z^T v_i\|^2 \leq 3a$ off probability $2ne^{-cd}$. Then $\|Y_i\| \leq \|Z^T v_i\| + \|Zu_i\| \leq \sqrt{3a} + 16\sqrt{3a/2}$.

(S2): $(\mathcal{A}Z)_i$ is Gaussian with variance $\leq \rho p_i/n \leq 3\rho a/(2n)$ ([Lemma 5.3\(b\)](#)); a union bound gives the first claim off probability $2n(2n)^{-3}$. The form $q_i(Z) = \langle Z, \mathbf{M}_i Z \rangle$, $\mathbf{M}_i Z = v_i v_i^T Z - Zu_i u_i^T$, has $\|\mathbf{M}_i\|_{\text{op}} \leq 1$ and $\|\mathbf{M}_i\|_{\mathbb{F}}^2 \leq 2(d + np_i^2) \leq 7d$; after the covariance factor $(C_{\rho}/n)^{1/2}$ these become $\leq 1/n$ and $\leq \sqrt{7d}/n$, and [Lemma A.1\(a\)](#) with $u = \eta'a$ gives failure probability $2n \exp(-c\eta'^2 d)$.

(S3): *Variance bookkeeping.* By the proof of [Lemma 5.8](#), for $i \neq k$ with $P_{ik}^2 \leq a/16$ the unfiltered variance of Y_{ik} is at least $a/(8n)$; at most 24 indices k have $P_{ik}^2 > a/16$. Row i of T_{e_j} is $\delta_{ij}e_j^T P + P_{ij}e_j^T - 2P_{ij}e_j^T P$, of squared norm at most $3(\delta_{ij}p_j + P_{ij}^2 + 4P_{ij}^2 p_j)$, so the base part removes from row i total variance $\frac{1}{n} \sum_j \|e_i^T T_{e_j}\|^2/p_j \leq \frac{6}{an} \cdot 5p_i \leq \frac{45}{n}$, which exceeds $\frac{a}{32n}$ at no more than $1440n/d$ positions. For $i \notin \mathfrak{B}$ the residual part removes at most $\delta_0 a$ from row i , hence more than $\frac{a}{32n}$ at no more than $32\delta_0 n$ positions. So for $d \geq 10^6$ (hence $n \geq 2 \cdot 10^6$, and the excluded indices number at most $25 + 0.0065n \leq 0.01n$) every $i \notin \mathfrak{B}$ has at least $0.99n$ indices $k \neq i$ with $\text{Var}(Y_{ik}) \geq \frac{a}{16n}$.

Small ball. Let $\phi : \mathbb{R} \rightarrow [0, 1]$ be C^1 , equal to 1 on $[-h, h]$, vanishing outside $[-2h, 2h]$, with $|\phi'| \leq 2/h$, and for $i \notin \mathfrak{B}$ let $F_i = \frac{1}{n} \sum_{k \neq i} \phi((P_{ik} + tY_{ik})/(t\sqrt{a/n}))$. By the density bound for a Gaussian of variance $\geq \frac{at^2}{16n}$, $\mathbb{E}F_i \leq 0.01 + \frac{4h}{\sqrt{2\pi/16}} \leq 0.01 + 7h$. As a function of the row $(Y_{ik})_k$, F_i has gradient norm at most $\frac{1}{n} \cdot \frac{2}{h} \cdot \sqrt{n/a} \cdot \sqrt{n} = 2/(h\sqrt{a})$; the row depends on Z through a linear map of norm ≤ 2 , and $Z = (C_{\rho}/n)^{1/2}g$ with g standard. So F_i is $4/(h\sqrt{d})$ -Lipschitz in g , and [Lemma A.1\(b\)](#) gives $\mathbb{P}(F_i > \mathbb{E}F_i + 0.01) \leq 2 \exp(-ch^2 d)$. Take $h = 1/500$, so that $\mathbb{E}F_i + 0.01 \leq 0.04$, and a union bound over i . On the complement, at most $0.04n$ indices $k \neq i$ have $|P_{ik} + tY_{ik}| < ht\sqrt{a/n}$, so at least $n - 1 - 0.04n \geq 0.95n$ have the reverse inequality.

(S4): Only second moments are needed, because $\mathfrak{B}$ is deterministic and bounded. Let $\Pi_{\mathfrak{B}}$ be the coordinate projection onto $\mathbb{R}^{\mathfrak{B}}$, $J_{\mathfrak{B}} = L_{\mathfrak{B}\mathfrak{B}} + at^2 I_{\mathfrak{B}}$ and $\omega_{ij} = J_{\mathfrak{B}}^{-1/2} \Pi_{\mathfrak{B}}(e_i - e_j)$, so $\|\omega_{ij}\|^2 \leq 2/(at^2)$ and $\sum_{i < j} P_{ij}^2 \omega_{ij} \omega_{ij}^T = J_{\mathfrak{B}}^{-1/2} L_{\mathfrak{B}\mathfrak{B}} J_{\mathfrak{B}}^{-1/2} \preceq I$. For x supported on $\mathfrak{B}$, $tx^T \mathcal{C}(Y)x = \langle J_{\mathfrak{B}}^{1/2} x, \Xi J_{\mathfrak{B}}^{1/2} x \rangle$ with $\Xi = \sum_{i < j} 2tP_{ij} Y_{ij} \omega_{ij} \omega_{ij}^T$, and $x^T J_{\mathfrak{B}} x = x^T Lx + at^2 \|x\|^2$. The coefficients $(Y_{ij})_{i < j}$ have covariance $\preceq \frac{2}{n} I$ (the map $Z \mapsto Y_Z$ is $\sqrt{2}$ times an isometry), so

$$\mathbb{E} \|\Xi\|_{\text{F}}^2 \leq \frac{2t^2}{n} \sum_{i < j} 4P_{ij}^2 \|\omega_{ij}\|^4 \leq \frac{16}{an} \text{tr}(J_{\mathfrak{B}}^{-1/2} L_{\mathfrak{B}\mathfrak{B}} J_{\mathfrak{B}}^{-1/2}) \leq \frac{16b}{d},$$

and Markov's inequality gives the first half off probability $16b/(d\eta^2)$. The principal $\mathfrak{B}$ -block of $\mathcal{L}(Y) - \mathbb{E}\mathcal{L}(Y)$ has diagonal entries $\sum_{j \neq i} (Y_{ij}^2 - \mathbb{E}Y_{ij}^2)$, a centred Gaussian quadratic form of variance at most $2 \cdot \frac{2}{n} \cdot \mathbb{E}\|Y_i\|^2 \leq \frac{12a}{n}$ (Lemma A.1(a), with $\mathbb{E}\|Y_i\|^2 \leq a(1-p_i) + (1-a)p_i \leq 3a$), and off-diagonal entries $-(Y_{ik}^2 - \mathbb{E}Y_{ik}^2)$ of variance at most $8/n^2$. Its expected squared Frobenius norm is at most $20b^2 a/n = 20b^2 a^2/d$, and Markov gives the second half off probability $20b^2/(d\eta^2)$.

Finally choose $A_{\eta'}$ so large that $d \geq A_{\eta'} \log(2n)$ implies $d \geq 10^6$ and all failure probabilities above are at most $\frac{1}{10}$; their sum is less than 1. $\square$

5.6 The drifted retraction

Fix $D \geq 12$ and $t \in (0, 1]$, set $\rho = D^2 t^2$, take a sample Z as in Lemma 5.9, and let m_* be as in Lemma 5.6. Define

$$W = Z + \frac{t}{2} m_*, \quad G = I + t^2 W^T W, \quad U_t = (U + tVW)G^{-1/2}, \quad P_t = U_t U_t^T.$$

Since $U^T V = 0$, $(U + tVW)^T (U + tVW) = G$ and U_t is Parseval.

Lemma 5.10. *There is an absolute C_E (depending only on the bounds in (S1) and on $D \geq 12$) such that:*

- (a) $\|U_t - U\|_{\text{F}}^2 \leq C_E t^2 d$;
- (b) every row of $\mathbf{E} = P_t - P - tY$ satisfies $\sum_j \mathbf{E}_{ij}^2 \leq at^2(D^{-2} + C_E t^2)$;
- (c) with $\|r\|_{\infty} \leq C_E at^3$,

$$\text{diag } P_t - a\mathbf{1} = (1 - t^2)(p - a\mathbf{1}) + 2tAZ - t^2 b_0 + t^2(q(Z) - \mathbb{E}q(Z)) + t^3 q(Z, m_*) + \frac{t^4}{4} q(m_*) + r;$$

$$(d) \|\text{diag } P_t - a\mathbf{1}\|_{\infty} \leq at^2 \left(\frac{\varepsilon}{t^2} + 6D\sqrt{\log(2n)/d} + \frac{12\varepsilon}{d} + \eta' + \frac{40}{D} + \frac{1}{2D^2} + C_E t \right).$$

Proof. By Lemma 5.6, $\frac{t}{2} \|m_*\|_{\text{op}} \leq \frac{t}{2} \|m_*\|_{\text{F}} \leq \sqrt{a/2}/D \leq \frac{1}{24}$, so with (S1): $\|W\|_{\text{op}} \leq 17$, $\|W\|_{\text{F}} \leq 17\sqrt{d}$, $\|W^T v_i\| \leq 2\sqrt{a}$, $\|W u_i\| \leq 17\sqrt{3a/2}$, and $\|(Y_{m_*})_i\|^2 \leq \|m_*\|_{\text{op}}^2 \leq 2a/\rho$. Hence $\|G^{-1} - I\|_{\text{op}}$ and $\|G^{-1/2} - I\|_{\text{op}}$ are at most $289t^2$, $\|U + tVW\|_{\text{op}} \leq 18$ and $\|u_i + tW^T v_i\| \leq 4\sqrt{a}$.

$$(a) \|U_t - U\|_{\text{F}} \leq \|U + tVW\|_{\text{F}} \|G^{-1/2} - I\|_{\text{op}} + t\|W\|_{\text{F}} \leq Ct\sqrt{d}.$$

(b) $\mathbf{E} = \frac{t^2}{2} Y_{m_*} + \mathbf{E}'$ with $\mathbf{E}' = P_t - P - tY_W = (U + tVW)(G^{-1} - I)(U + tVW)^T + t^2 VWW^T V^T$, whose rows have norm at most $Ct^2 \sqrt{a}$. The rows of $\frac{t^2}{2} Y_{m_*}$ have squared norm at most $\frac{t^4}{4} \cdot \frac{2a}{\rho} = \frac{at^2}{2D^2}$. Use $(\alpha + \beta)^2 \leq 2\alpha^2 + 2\beta^2$.

(c) Let $\xi_i = u_i + tW^T v_i$, the i th row of $U + tVW$. Inserting $G^{-1} = I - t^2 W^T W + t^4 (W^T W)^2 G^{-1}$ into $(P_t)_{ii} = \xi_i^T G^{-1} \xi_i$ and using $u_i^T W^T v_i = (\mathcal{A}W)_i$,

$$(P_t)_{ii} = p_i + 2t(\mathcal{A}W)_i + t^2 q_i(W) - 2t^3 \langle W u_i, W W^T v_i \rangle - t^4 \|W W^T v_i\|^2 + t^4 \xi_i^T (W^T W)^2 G^{-1} \xi_i,$$

and the last three terms are $O(at^3)$. Now $2tAW = 2tAZ + t^2Am_*$, $q(W) = q(Z) + tq(Z, m_*) + \frac{t^2}{4}q(m_*)$, and $\mathbb{E}q(Z) = a\mathbf{1} - p - b_0 - Am_*$ by Lemma 5.6; the terms $\pm t^2Am_*$ cancel.

(d) In (c): $(1 - t^2)\|p - a\mathbf{1}\|_\infty \leq \varepsilon a$; $2t\|AZ\|_\infty \leq 6Dt\sqrt{a \log(2n)/n}$ by (S2), which is $6Dat^2\sqrt{\log(2n)/d}$; $t^2\|b_0\|_\infty \leq 12\varepsilon t^2/n$; the fluctuation is at most $\eta'at^2$; by (5.4) and (S1), $t^3|q_i(Z, m_*)| \leq t^3\sqrt{800a}\sqrt{2a/\rho} = 40at^2/D$ and $\frac{t^4}{4}|q_i(m_*)| \leq \frac{t^4}{4} \cdot \frac{2a}{\rho} = \frac{at^2}{2D^2}$. $\square$

Lemma 5.11 (Graph of the seed). *There are absolute $t_* \in (0, 1]$ and C_{core} such that the following holds. Let $\eta' \leq \frac{1}{32}$, $d \geq A_{\eta'} \log(2n)$, $D \geq 13/h$, $b \leq n/10$ and $t \leq t_*$, and let Z satisfy (S1)–(S4). Then $H(L_{P_t}) \leq C_{\text{core}}/(at^2)$.*

Proof. Core. In a row $i \notin \mathfrak{B}$, by Lemma 5.10(b), at most $at^2(D^{-2} + C_E t^2)/(\frac{h^2}{4} \frac{at^2}{n}) \leq 0.05n$ indices have $|\mathbf{E}_{ij}| \geq \frac{h}{2}t\sqrt{a/n}$, once $D \geq 13/h$ and $t^2 \leq h^2/(160C_E)$. With (S3), every $i \notin \mathfrak{B}$ has at least $0.9n$ indices $j \neq i$ with $(P_t)_{ij}^2 \geq \frac{h^2}{4} \frac{at^2}{n}$.

Block. For x supported on $\mathfrak{B}$, Lemma 5.8, $x^T K x \geq (n - b)\|x\|^2$ and (S4) give

$$x^T \mathcal{L}(P + tY)x \geq \left(1 - \eta' - t^2\left(\frac{2}{n} + \frac{8}{d}\right) - \frac{8}{D^2}\right)x^T Lx + at^2\left(\frac{1}{4}\left(1 - \frac{b}{n}\right) - 2\eta'\right)\|x\|^2 \geq \frac{at^2}{8}\|x\|^2,$$

since the first coefficient is positive and $\frac{1}{4} \cdot \frac{9}{10} - \frac{1}{16} \geq \frac{1}{8}$. By (5.1) and Lemma 5.10(b), $L_{P_t} = \mathcal{L}(P + tY + \mathbf{E}) \succeq \frac{1}{2}\mathcal{L}(P + tY) - 2at^2(D^{-2} + C_E t^2)I$, which is at least $\frac{at^2}{32}$ on vectors supported on $\mathfrak{B}$ once $t^2 \leq 1/(128C_E)$.

Barrier. Lemma 2.10 with $\mathcal{B} = \mathfrak{B}$, part (i) with $\vartheta = 0.4$, $\gamma = h^2 at^2/4$, and part (ii) with $\lambda = at^2/32$, gives $H(L_{P_t}) \leq \frac{1+b}{0.1h^2 at^2} + \frac{32\sqrt{b}}{at^2}$. Put $C_{\text{core}} = 10(1 + b_{\max})/h^2 + 32\sqrt{b_{\max}}$. $\square$

5.7 Proof of Theorem 5.1

Constants. The numbers $\delta_0, b_{\max}, h, C_E, t_*, C_{\text{core}}$ are absolute and do not depend on D (for $D \geq 12$). Put $\Theta = \max\{2, C_{\text{core}}, C_E\}$ and choose, in this order: $D \geq \max\{13/h, 500\Theta\}$, so that $\frac{40}{D} + \frac{1}{2D^2} \leq \frac{1}{12\Theta}$; $K_0 = 12\Theta$ and $\eta' = \min\{\frac{1}{32}, \frac{1}{12\Theta}\}$; $A \geq A_{\eta'}$ so large that $d \geq A \log(2n)$ implies $6D\sqrt{\log(2n)/d} \leq \frac{1}{12\Theta}$ and $b_{\max} \leq n/10$; and $\varepsilon_0 \leq \frac{1}{2}$ so that $t = \sqrt{K_0\varepsilon}$ satisfies $t \leq t_*$, $C_E t \leq \frac{1}{12\Theta}$ and $12\varepsilon \leq \frac{1}{12\Theta}$ for $\varepsilon \leq \varepsilon_0$. None of these choices depends on n, d, ε or U .

Seed. With $t^2 = K_0\varepsilon$, Lemma 5.10(d) bounds the diagonal error of P_t by $at^2 \cdot \frac{6}{12\Theta} = \frac{at^2}{2\Theta}$; Lemma 5.11 bounds the barrier constant by $\Theta/(at^2)$; and Lemma 5.10(a) bounds $\|U_t - U\|_{\text{F}}^2$ by $\Theta t^2 d$. So U_t is a (Θ, t) -seed for U , and Corollary 2.13 gives an ENP frame $\hat{U}$ with $\|\hat{U} - U\|_{\text{F}}^2 \leq 3\Theta t^2 d = 36\Theta^2 \varepsilon d$. $\square$

Remark 5.12. The drift keeps the spectral input local. The bias $t^2 b_*$ could also be removed after the perturbation, for instance by a second scaling; since $\langle x, b_* \rangle^2 \leq \frac{2a}{\rho} x^T Lx$ this is possible, but it requires two-sided control of L_{P_t} on all of $\mathbf{1}^\perp$. With the drift, the only spectral input is the gap on the bounded deterministic block $\mathfrak{B}$, which needs second moments only.

6 Assembly

We use the two seeds through the following equal-norm formulation.

Proposition 6.1. *There is an absolute constant C_0 such that every $X \in \mathbb{R}^{n \times d}$ with $1 \leq d$, $2d \leq n$ and $\|x_i\|^2 = a$ for all i admits an ENP frame W with*

$$\|X - W\|_{\text{F}}^2 \leq C_0 d \|X^T X - I\|_{\text{op}}.$$

Proof. Let $\eta = \|X^T X - I\|_{\text{op}}$; if $\eta = 0$ take $W = X$. Let A, ε_0, C_m be the constants of [Theorem 5.1](#) and B, c_*, C_* those of [Theorem 4.1](#), and fix an integer $d_0 \geq 2$ with $A \log(2Bd^2) \leq d$ for all $d \geq d_0$.

Case $d < d_0$. [Lemma 3.1](#) gives $\|X - W\|_{\text{F}}^2 \leq \eta d(d-1) \leq d_0 \eta d$.

Case $d \geq d_0$, $n \geq Bd^2$. If $\eta \leq c_*$ use [Theorem 4.1](#); otherwise [Corollary 3.2](#) gives cost $4d \leq (4/c_*)\eta d$.

Case $d \geq d_0$, $n < Bd^2$. Then $A \log(2n) \leq d$. If $\eta \leq \varepsilon_0/4$, let $U = X(X^T X)^{-1/2}$. The singular values of X lie in $[\sqrt{1-\eta}, \sqrt{1+\eta}]$, so $\|X - U\|_{\text{F}}^2 \leq d\eta^2$, and $\|u_i\|^2 = x_i^T (X^T X)^{-1} x_i \in [a/(1+\eta), a/(1-\eta)]$, so $||u_i\|^2 - a| \leq 2\eta a$. [Theorem 5.1](#) with $\varepsilon = 2\eta$ gives an ENP W with $\|U - W\|_{\text{F}}^2 \leq 2C_m \eta d$, and $\|X - W\|_{\text{F}}^2 \leq 2d\eta^2 + 4C_m \eta d$. If $\eta > \varepsilon_0/4$ use the trivial cost $4d \leq (16/\varepsilon_0)\eta d$. Altogether $C_0 = \max\{d_0, C_*, 4/c_*, 2 + 4C_m, 16/\varepsilon_0\}$ works. $\square$

Proof of [Theorem 1.2](#). Replacing (P, Q) by $(I - P, I - Q)$ preserves β , ranks of complements and distances, so we may assume $d \leq n/2$; the case $d = 0$ is trivial. Write $P = UU^T$ with $U^T U = I$, $p_i = \|u_i\|^2$ and $\delta = \beta/a$. If $\delta \geq \frac{1}{2}$, any ENP W ([Corollary 3.2](#)) gives $\|P - WW^T\|_{\text{F}}^2 \leq 2d \leq 4n\beta$. Otherwise let $x_i = \sqrt{a} u_i / \|u_i\|$. Then $\|X - U\|_{\text{F}}^2 = \sum_i (\sqrt{p_i} - \sqrt{a})^2 \leq \sum_i (p_i - a)^2 / a \leq \delta^2 d$ and $X^T X = U^T \text{Diag}(a/p) U$, so $\|X^T X - I\|_{\text{op}} \leq \max_i |a/p_i - 1| \leq \delta/(1-\delta) \leq 2\delta$. [Proposition 6.1](#) gives an ENP W with $\|U - W\|_{\text{F}}^2 \leq 2\delta^2 d + 4C_0 \delta d \leq (1 + 4C_0)n\beta$, using $\delta d = n\beta$. Take $Q = WW^T$; by [Lemma 2.1](#), $\|P - Q\|_{\text{F}}^2 \leq 2(1 + 4C_0)n\beta$. So [Theorem 1.2](#) holds with $C_P = \max\{4, 2 + 8C_0\}$. $\square$

Proof of [Theorem 1.1](#). If $\varepsilon \geq \frac{1}{2}$, an ENP W gives $\|X - W\|_{\text{F}}^2 \leq 2\|X\|_{\text{F}}^2 + 2d \leq 6d \leq 12\varepsilon d$, since $\|X\|_{\text{F}}^2 = \text{tr } X^T X \leq (1 + \varepsilon)d$. Let $\varepsilon < \frac{1}{2}$ and $U = X(X^T X)^{-1/2}$, $P = UU^T$. As above $\|X - U\|_{\text{F}}^2 \leq \varepsilon^2 d$ and $P_{ii} = x_i^T (X^T X)^{-1} x_i \in [\frac{1-\varepsilon}{1+\varepsilon}a, \frac{1+\varepsilon}{1-\varepsilon}a]$, so $\beta = \max_i |P_{ii} - a| \leq 4\varepsilon a$. [Theorem 1.2](#) gives Q with $\|P - Q\|_{\text{F}}^2 \leq 4C_P \varepsilon d$, and by [Lemma 2.1](#) there is an isometry W spanning Q —hence ENP—with $\|U - W\|_{\text{F}}^2 \leq \|P - Q\|_{\text{F}}^2$. Then $\|X - W\|_{\text{F}}^2 \leq 2\varepsilon^2 d + 8C_P \varepsilon d$, so [Theorem 1.1](#) holds with $C = \max\{12, 1 + 8C_P\}$. $\square$

A Gaussian tools

Lemma A.1. *The following hold with absolute constants.*

(a) (Quadratic forms.) For g standard Gaussian and $\mathbf{A}$ symmetric, $\text{Var}(g^T \mathbf{A} g) = 2\|\mathbf{A}\|_{\text{F}}^2$ and

$$\mathbb{P}\{|g^T \mathbf{A} g - \text{tr } \mathbf{A}| > u\} \leq 2 \exp\left[-\frac{1}{8} \min\left\{\frac{u^2}{\|\mathbf{A}\|_{\text{F}}^2}, \frac{u}{\|\mathbf{A}\|_{\text{op}}}\right\}\right].$$

For a Gaussian vector $\zeta = \Sigma^{1/2} g$, apply this to $\Sigma^{1/2} \mathbf{A} \Sigma^{1/2}$, whose norms are at most $\|\Sigma\|_{\text{op}}$ times those of $\mathbf{A}$.

(b) (Lipschitz functions.) If F is C^1 with $\|\nabla F\| \leq \ell$ and g is standard Gaussian, then for $u \geq 0$ $\mathbb{P}\{|F(g) - \mathbb{E}F(g)| > u\} \leq 2 \exp(-2u^2/(\pi^2 \ell^2))$.

(c) (Nets.) If Z is an $r \times s$ centred Gaussian matrix with $\text{Cov}(\text{vec } Z) \preceq \sigma^2 I$, then for $u \geq 0$ $\mathbb{P}\{\|Z\|_{\text{op}} > u\} \leq 2 \cdot 5^{r+s} \exp(-u^2/(32\sigma^2))$.

Proof. (a) Diagonalise $\mathbf{A}$ with eigenvalues λ_j . For $|s| \leq 1/(4\|\mathbf{A}\|_{\text{op}})$, $\log \mathbb{E} e^{s(g^T \mathbf{A} g - \text{tr } \mathbf{A})} = \sum_j [-s\lambda_j - \frac{1}{2} \log(1 - 2s\lambda_j)] \leq 2s^2 \|\mathbf{A}\|_{\text{F}}^2$, since $-x - \frac{1}{2} \log(1 - 2x) = \sum_{k \geq 2} (2x)^k / (2k) \leq 2x^2$ for $|x| \leq \frac{1}{4}$. The exponential Markov inequality with $s = \min\{u/(4\|\mathbf{A}\|_{\text{F}}^2), 1/(4\|\mathbf{A}\|_{\text{op}})\}$, applied to both signs, gives the tail. The variance follows from $\mathbb{E} g_j^4 = 3$.

(b) Let g' be an independent copy, $g_\theta = g \cos \theta + g' \sin \theta$ and $\dot{g}_\theta = -g \sin \theta + g' \cos \theta$; for each θ the pair $(g_\theta, \dot{g}_\theta)$ is standard Gaussian. By the fundamental theorem of calculus and Jensen on $[0, \pi/2]$,

$$\mathbb{E} e^{s(F(g') - F(g))} \leq \frac{2}{\pi} \int_0^{\pi/2} \mathbb{E} \exp \left[\frac{\pi s}{2} \langle \nabla F(g_\theta), \dot{g}_\theta \rangle \right] d\theta \leq e^{\pi^2 s^2 \ell^2 / 8}.$$

Jensen in g' gives $\mathbb{E} e^{s(F(g) - \mathbb{E} F)} \leq e^{\pi^2 s^2 \ell^2 / 8}$; optimise in s . All expectations are finite since $|F(x)| \leq |F(0)| + \ell \|x\|$, and Fubini applies to the θ -integral.

(c) Let $\mathcal{N}_1, \mathcal{N}_2$ be $\frac{1}{2}$ -nets of the unit spheres of $\mathbb{R}^r$ and $\mathbb{R}^s$ with at most $5^r, 5^s$ points. For a vector w , a net point y' within $\frac{1}{2}$ of $w/\|w\|$ has $\langle w, y' \rangle \geq \frac{1}{2} \|w\|$; applying this twice gives $\|Z\|_{\text{op}} \leq 4 \max_{x \in \mathcal{N}_1, y \in \mathcal{N}_2} |x^T Z y|$. Each $x^T Z y$ is Gaussian with variance $\leq \sigma^2$; a union bound finishes. $\square$

Covariance domination. If a Gaussian vector ξ has covariance $\Gamma \preceq \gamma I$ and (M_α) are matrices, then $\mathbb{E} \|\sum_\alpha \xi_\alpha M_\alpha\|_F^2 = \text{tr}(\Gamma G) \leq \gamma \sum_\alpha \|M_\alpha\|_F^2$, where $G \succeq 0$ is the Gram matrix of the M_α . Correlated Gaussians enter only as $\Sigma^{1/2} g$: in quadratic forms through the last sentence of (a) (Lemma 4.3, Lemma 5.9), in Lipschitz functions through (b), in operator norms through (c), and in Frobenius norms of linear combinations through this identity (Lemma 5.9(S4)).

B Formalisation

Theorem 1.1 is formalised in Lean 4 with Mathlib; the development accompanies this paper (directory `lean/`). The formal statement is

```
def SharpPaulsenBound : Prop :=
  ∃ C : ℝ, 0 < C ∧
  ∀ (n d : ℕ), 0 < d → d ≤ n →
  ∀ (ε : ℝ) (U : Frame n d), 0 < ε → ε < 1 →
  IsNearlyEqualNormParseval ε U →
  ∃ W : Frame n d, IsEqualNormParseval W ∧
  sqDistance U W ≤ C * ε * (d : ℝ)
```

where `Frame n d` is the type of real $n \times d$ matrices, Parsevalness is $U^T U = 1$, the near-Parseval condition is stated through the quadratic form $x \mapsto \|Ux\|^2$, the row-norm conditions refer to d/n as a real quotient, and `sqDistance` is the unnormalised squared Frobenius distance. The projection form (Theorem 1.2) is derived for arbitrary real symmetric idempotent matrices of rank d . Both statements are also restated in plain Mathlib vocabulary (positive semidefiniteness, dot products, traces, `Matrix.rank`), and the restatements are derived from them.

Structure. The directory `Paulsen/Paper` (about 16,100 lines) contains, for every numbered definition, lemma, theorem, corollary and proposition of this paper, for every numbered display that makes a claim, and for every remark that makes a quantitative claim, a Lean statement with *exactly the constants of the text*. Each declaration quotes the corresponding T_EX in its documentation. The explicit claims made inside the longer proofs (the constant choices in the proofs of Theorems 2.8, 4.1 and 5.1, Lemmas 5.9 to 5.11, and Proposition 6.1, and of Theorems 1.1 and 1.2) are stated as separate lemmas, so the constants used inside proofs are pinned down as well. All of these statements are proved. The two main theorems are derived from them along the route of the text: Theorem 1.1 from Theorem 1.2, which comes from Proposition 6.1, which in turn comes from Lemma 3.1 and Theorems 4.1 and 5.1. These rest on the lemmas of Sections 2, 4

and 5 and [Appendix A](#). The remaining files form a supporting library (frames, projections, the log-determinant potential, Gaussian measures, the noises of [Sections 4](#) and [5](#), the Hamilton–Moitra argument).

paper	Lean (Paulsen/Paper/)
Section 1 (Remark 1.3)	Intro
Section 2	Toolbox (helpers <code>ToolboxAux*</code>)
Section 3	Bounded
Section 4	ManyRow, ManyRowProof (helpers <code>ManyRowAux*</code>)
Section 5 , up to Lemma 5.8	Moderate (helpers <code>ModerateAux*</code>)
Lemma 5.9	ModerateSample (helpers <code>SampleAux*</code>)
Lemmas 5.10 and 5.11 and Theorem 5.1	ModerateSeed (helpers <code>SeedAux*</code>)
Section 6 , Theorems 1.1 and 1.2	Assembly
Appendix A	Gaussian

Encoding. Operator norms and Loewner inequalities are expressed through quadratic forms or the norm of the associated linear map. In [Section 5](#) the horizontal noise is represented in ambient coordinates, $H = VZ \in \mathbb{R}^{n \times d}$ with $U^T H = 0$. Gaussian vectors are images of the standard Gaussian measure under linear maps, “with positive probability” means that the set of good samples has positive measure, and expectations are Lebesgue integrals. Where the text writes $\max_i$ or $\|\cdot\|_\infty$, the formal statement takes an arbitrary upper bound; all conclusions are monotone, so this is equivalent.

Constants. Explicit constants appear verbatim. Constants that the text only asserts to exist are existential in the formal statements, and the proofs exhibit admissible values:

- $C_{\mathcal{M}} = 192$ and $h = \frac{1}{300}$, $t_0 = \frac{1}{30}$ in [Section 4](#);
- $A_{\eta'} = 10^{12}/\eta'^2$ in [Lemma 5.9](#);
- $C_E = 2 \cdot 10^9$ in [Lemma 5.10](#).

The formalisation found no false constant. It made a few implicit hypotheses explicit, and these are now stated in the text: $u \geq 0$ in [Lemma A.1](#)(b),(c); $d^2 \leq n$ in [Lemma 4.2](#); and $\varepsilon_0 \leq \frac{1}{2}$ in [Theorem 5.1](#).

Checks. The project builds with Lean 4.32.0 and Mathlib v4.32.0. `Audit.lean` checks four things:

- the final theorems have the target types;
- every theorem of the project depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`, so that no statement is left unproved;
- every lemma, theorem, corollary, proposition and numbered claim of [Paulsen/Paper](#) lies in the dependency cone of the two final theorems;
- the proofs use none of the library lemmas whose constants are weaker than those of the text.

The exceptions to the third check are two existential forms whose explicit versions are used, the eigenvalue facts [\(5.2\)](#), and the definitional restatement of [\(1.1\)](#). The remarks are proved as standalone statements. Every module was also replayed in the kernel with `leanchecker`.

Contributions and acknowledgements

K. Hänni and H. Eberhard proposed the overall strategy: perturb by a random tangent of size $\sqrt{\varepsilon}$ to create a well-connected squared-Gram graph, and then balance by a diagonal scaling controlled in ℓ_∞ . Astra developed a complete proof along these lines, including both seeds and the reduction to them, together with the Lean library on which the formalisation rests. Claude contributed the cancelling drift in the moderate-row seed, the centred remainder estimate that makes the many-row seed work at every error level, and the barrier-constant formulation of balancing; wrote this text; and completed the formalisation of the proof given here.

References

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